

D2.6

Requirement elicitation and specification for antifragile urban mobility

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Authors	Cristina De Nardi, Afrouz Ghaemi, Yacine Rezgui, Ali Ghoroghi, Andre Hodorog
Reviewers	Thomas Beach, Linghang Sun

Abstract

This deliverable specifies the requirements for the Simulator for Urban Mobility Antifragility (SUMA / KER5), the decision-support software environment at the core of the AntifragiCity project. The report consolidates requirements derived from (a) a critical review of the literature, (b) project partners, including our 3 municipalities, namely Bratislava, Larissa, and Odesa, using a dedicated consultation, involving a proforma and follow-on workshop, and (c) stakeholder engagement, and translates them into an implementable specification. To validate and prioritise the requirement set, we apply a Delphi-inspired expert consultation approach adapted for project timelines: a structured single-round scoring of pre-defined requirement statements, supported by transparent consensus

rules. Experts rate each requirement on a 7-point Likert scale, and consensus is achieved using an Interquartile Range (IQR)-based threshold to classify requirements into high-, moderate-, or low-consensus categories. The resulting requirement set (47 statements) and its prioritisation provide a traceable basis for SUMA design, implementation planning, and iterative refinement through WP5 agile feedback loops.

Keywords

SUMA; requirements elicitation; decision support; urban mobility resilience; antifragility; Delphi-inspired expert consultation; consensus (IQR); prioritisation; traceability; WP5 integration.

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R *

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PU	Public, fully open. e.g., website	✓
CL	Classified information as referred to in Commission Decision 2001/844/EC	
CO	Confidential to AntifragiCity project and Commission Services	

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R: document, report (excluding periodic and final reports).

DEM: demonstrator, pilot, prototype, plan designs.



DEC: websites, patent filings, press and media actions, videos, etc.

OTHER: software, technical diagrams, etc.

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Abbreviations

AFC	AntifragiCity
API	Application Programming Interface
DR	Data Requirement(s)
EU	European Union
FR	Functional Requirement(s)
GA	General Assembly
GDPR	General Data Protection Regulation
GR	Governance Requirement(s)
IQR	Interquartile Range
IR	Interoperability/Integration Requirement(s)
KPI	Key Performance Indicator(s)
KER	Key Exploitable Result
ML	Machine Learning
NFR	Non-Functional Requirement(s)
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
SUMA	Simulator for Urban Mobility Antifragility
SUMO	Simulation of Urban Mobility
TR	Traceability Requirement(s)
UI	User Interface
UR	Usability Requirement(s)
WP	Work Package

1 Executive summary

This deliverable specifies the **requirements** for the **Simulator for Urban Mobility Antifragility (SUMA)**, forming the AntifragiCity 5th Key Exploitable Result (KER5). The decision-support software environment is at the heart of AntifragiCity. WP5 (The Simulator for Urban Mobility Antifragility – SUMA) will assemble results from WP2 (Defining urban mobility crises, acceptance and antifragile framework), including the event ontology, WP3 (Defining urban mobility crises, acceptance and antifragile framework), and WP4 (Antifragile traffic control in urban road networks). It will integrate these results into a user-friendly environment compatible with existing transportation simulation software and standalone use. SUMA's objective is to provide an **urban mobility API** enabling stakeholders (e.g., transport planners and city officials) to interface their tools, assess fragility in a single process and explore interventions that improve resilience by implementing antifragile attributes, including learning from past disturbances and stressors.

This report (i) defines the requirement elicitation approach, (ii) states stakeholder and user needs, (iii) formalises **functional and non-functional requirements**, and (iv) sets traceability and acceptance criteria aligned with WP5 deliverables (Core SUMA API specification; implementation & integration; role-based UI; agile feedback loop).

A core element of this deliverable is an expert consensus process (Delphi-style) to validate and prioritise requirements, leveraging key Delphi principles including independent judgement, structured assessment, and statistical aggregation of expert ratings.

2 Introduction

Increasing climate variability, rapid urbanisation, and cascading infrastructure failures are intensifying the exposure of cities to disruptive events, challenging the capacity of traditional resilience-based approaches to ensure long-term urban performance [1]. While resilience focuses on resistance and recovery, it often assumes a return to pre-disturbance conditions, potentially limiting the ability of urban systems to adapt to deep uncertainty [2]. In response, the concept of antifragility has emerged as a promising paradigm, describing systems that not only withstand shocks but learn, adapt, and improve when exposed to instability [3]. Emerging studies highlight that crises can act as catalysts for innovation and transformation, supporting planning and governance models that leverage disruption to enhance long-term urban performance and sustainability [4]. Consequently, there is growing interest in operationalising antifragility within planning frameworks [5]. System responses can be interpreted along a spectrum ranging from fragility to robustness, resilience, and ultimately antifragility, reflecting progressively greater capacity to leverage stressors and anticipate volatile conditions [6]. Importantly, antifragility is grounded in nonlinear system behaviour, whereby convex input-output responses enable performance gains when exposed to perturbations [6]. Such behaviour is frequently associated with heterogeneity, multi-scale interactions, and redundant overcompensation mechanisms that enhance adaptive capacity [7]. Moreover, efficient connectivity and information flow support adaptive reconfiguration, while stability itself may depend on a system's ability to modify its behaviour in response to external stimuli [6]. Consequently, antifragility is increasingly recognised as a design principle for complex engineered and socio-technical systems, supporting strategies that harness uncertainty to strengthen long-term adaptability and performance.

Mobility networks are particularly suited to antifragile design due to their high connectivity, dynamic demand patterns, and dependence on real-time information flows. Recent work highlights that transportation systems should not only resist disruptions but develop the capacity to reorganise and enhance performance under uncertain conditions [8]. This perspective aligns with growing interest in adaptive and learning-based mobility management strategies capable of responding to interconnected shocks such as extreme weather, pandemics, or infrastructure failures [1].

Despite these advances, the operational translation of antifragility into mobility planning remains limited. Existing studies either conceptualise antifragility at a systems level or rely on resilience-oriented transport models, leaving a methodological gap between theory and implementation. Consequently, there is a need for scalable methodologies that integrate adaptive capacity, system learning, and performance evolution into the design and governance of urban mobility networks. Addressing this gap, the present study advances a framework for embedding antifragile principles into mobility systems, aiming to support infrastructures that not only endure disruptions but leverage them to enhance long-term functionality and sustainability.

Building on the growing recognition that antifragility must be delivered through implementable system capabilities rather than remain a purely conceptual construct, this deliverable adopts a structured requirements elicitation methodology designed to translate theoretical principles into actionable design specifications.

This work aims to embed antifragility within urban mobility planning by translating conceptual principles into a structured and implementable requirements framework for the Simulator for Urban Mobility Antifragility (SUMA). Building on the theoretical and methodological foundations, the deliverable is organised to progressively bridge conceptual foundations, methodological design, and practical specification. Section 4 introduces the methodology, which involves multi-source requirement elicitation combining ontology development, systematic literature synthesis, project

documentation analysis, stakeholder and pilot input, and a Delphi-inspired expert consultation used to validate and prioritise the resulting requirement set. Section 5 presents the conceptual foundation underpinning requirement elicitation, outlining the ontology framework that defines the system scope and reviewing Delphi and Delphi-inspired approaches that inform the expert consultation design. Section 6 presents the results of the Delphi-inspired expert consultation, including category refinement, expert panel composition, and the analysis of agreement used to validate and prioritise the requirement set. Section 7 reports the requirement specification framework, outlining the structure used to organise the prioritised requirements, including the questionnaire basis, requirement categories, and cross-cutting thematic domains. Section 7 presents also the structured requirements catalogue for SUMA across functional, data, interoperability, usability, and non-functional dimensions, followed by Section 8, which details requirements management and traceability. Section 8 outlines assumptions, constraints, and open issues. Section 9 discusses the implications of the findings, and Section 10 concludes the deliverable by summarising contributions and outlining directions for implementation and iterative refinement within WP5. Section 12 provides the references supporting the conceptual and methodological framework, and Section 12 presents the appendices, including detailed Delphi materials, instruments, and supplementary methodological documentation.

3 Related work

The development of requirements for urban mobility systems has evolved significantly over the past three decades, moving from predominantly infrastructure-centred transport planning toward systemic, user-centred, data-driven and resilience-oriented frameworks. The literature reveals several research threads, including transport planning theory, systems engineering, resilience and sustainability science, smart city research, and participatory design, that have been considered to inform the right requirement elicitation approach for complex urban mobility systems.

Early work in urban transport planning focused primarily on forecasting travel demand and optimizing network capacity. The traditional “predict and provide” paradigm relied heavily on four-step travel demand models (trip generation, distribution, modal split, assignment) and sought to derive system requirements from projected traffic volumes [9]. Requirements were expressed in terms of infrastructure capacity, level of service, and network efficiency. However, critiques in the 1990s and 2000s highlighted the limitations of this supply-oriented framing, particularly its failure to address sustainability, social equity, and behavioural change [10]. This shift prompted the development of “sustainable mobility” frameworks, which reframed requirements around accessibility, emissions reduction, and modal integration rather than traffic throughput.

The accessibility turn represents a key conceptual foundation for requirement development. Scholars such as Geurs and van Wee [11] argued that mobility systems should be assessed according to their ability to provide access to opportunities, rather than simply enabling movement. From a requirement engineering perspective, this reframing implies that system requirements must capture spatial equity, proximity to services, multimodal connectivity, and temporal reliability. Accessibility metrics, including cumulative opportunity measures and gravity-based models, began to inform transport performance indicators and policy targets, thereby shaping high-level system requirements.

Parallel to these developments, systems engineering methodologies were increasingly applied to transportation. Complex adaptive systems theory conceptualized urban mobility as a network of

interacting subsystems, including physical infrastructure, vehicles, users, governance structures, and digital platforms [12], [13]. This systems perspective emphasized interdependencies, feedback loops, and emergent behaviour. Requirements capture methodologies consequently began incorporating stakeholder analysis, functional decomposition, scenario modelling, and iterative validation processes [14]. In transport contexts, these approaches have been used to derive requirements for intelligent transportation systems (ITS), integrated ticketing platforms, and multimodal coordination frameworks.

The rise of intelligent and connected mobility further transformed requirement development. The deployment of ITS, traffic management systems, and real-time information platforms introduced new technical and socio-technical requirements, including interoperability, cybersecurity, data governance, and user privacy [15]. Research on smart cities [16], [17] this lens, positioning urban mobility as a core component of data-driven urban systems. Requirements thus evolved to include open data standards, platform integration, real-time analytics capability, and adaptive control mechanisms. The literature emphasizes that digital infrastructures must be designed with modularity and scalability to accommodate technological change and uncertain future demand.

Resilience theory has become particularly influential in the past fifteen years. Drawing from ecological resilience [18], scholars have conceptualized urban mobility systems as socio-technical systems exposed to shocks and stresses, including climate events, pandemics, infrastructure failure, and economic disruption [19]. Resilience-oriented requirement capture typically identifies capabilities such as redundancy, diversity of modes, network modularity, rapid recovery, and adaptive governance. Studies following major disruptions, for example, post-Hurricane Sandy transport analyses or COVID-19 mobility shifts, have emphasized the importance of flexibility, distributed services, and demand-responsive operations [3], [20]. These insights inform requirements for multimodal integration, distributed energy systems for electrified fleets, and contingency planning mechanisms.

Closely related is the emerging concept of antifragility, introduced by Taleb [3], [21] which goes beyond resilience by suggesting that systems can improve through volatility and stress. Although antifragility has been less used in transport planning, adjacent literatures provide building blocks for requirement formulation. Research on adaptive governance [21] emphasizes institutional learning and iterative policy adjustment. Work on modular network design [22] demonstrates how loosely coupled subsystems can enhance adaptability. In mobility contexts, this translates into requirements for modular infrastructure, open standards, experimentation zones (e.g., urban living labs), and dynamic pricing mechanisms that learn from behavioural feedback.

Participatory planning and co-creation processes also shape requirement capture methodologies. Arnstein's ladder of participation [23] remains foundational, but contemporary mobility research emphasizes co-design, citizen engagement platforms, and deliberative governance. Studies show that participatory processes improve legitimacy, user acceptance, and alignment with local needs [24]. From a requirement engineering standpoint, this implies systematic stakeholder mapping, inclusive consultation processes, and iterative validation of functional and non-functional requirements. Human-centred design approaches, including service design methods, are increasingly used to translate user journeys and lived experiences into system specifications.

Sustainability frameworks further influence requirement development. International policy instruments such as the Sustainable Development Goals (SDGs), particularly SDG 11 (sustainable cities and communities), and European policy frameworks like the Sustainable Urban Mobility Plan (SUMP) guidelines provide structured criteria for mobility systems [25]. These frameworks integrate environmental, social, and economic dimensions, requiring emissions reduction, inclusivity, affordability, and safety to be embedded as core system requirements. Lifecycle assessment

methodologies and carbon budgeting approaches increasingly inform transport infrastructure requirements, particularly in light of decarbonisation targets aligned with the Paris Agreement.

The electrification of mobility and the transition to shared and autonomous systems introduce additional layers of complexity. Research on electric vehicle (EV) infrastructure planning highlights requirements related to charging network optimization, grid integration, and demand management [26]. Autonomous vehicle studies emphasize safety validation, regulatory compliance, human-machine interaction, and ethical governance [27]. Shared mobility research underscores platform governance, data interoperability, and equitable access [28]. These domains collectively demonstrate that requirement capture must address cross-sectoral integration, particularly between transport, energy, telecommunications, and urban planning.

Methodologically, the literature highlights several structured approaches to requirement capture in complex mobility systems. These include scenario planning to address uncertainty [29], back casting from sustainability targets [30], systems dynamics modelling to explore feedback loops [31] and multi-criteria decision analysis to evaluate trade-offs among competing objectives [26], [32]. Increasingly, digital twins and simulation environments are used to test functional requirements under diverse stress conditions before physical implementation.

In synthesis, existing research indicates that effective requirement capture for urban mobility systems must integrate: (1) accessibility and equity principles; (2) systems engineering and modular design; (3) digital interoperability and data governance; (4) sustainability and decarbonisation objectives; (5) resilience and adaptive capacity; and (6) participatory governance mechanisms. The literature also reveals a progression from static, infrastructure-centred requirements toward dynamic, adaptive and learning-oriented specifications capable of responding to uncertainty.

An antifragile urban mobility system therefore builds upon established resilience and sustainability frameworks but extends them by embedding mechanisms for learning, experimentation, modular reconfiguration, and performance improvement under stress. Requirement capture in this context must not only specify robustness and redundancy but also define processes for iterative feedback, data-driven adjustment, and institutional adaptability. By integrating systems thinking, participatory co-design, sustainability science, and adaptive governance, contemporary research provides a robust conceptual foundation for articulating the technical, social, and institutional requirements necessary to guide the development of antifragile urban mobility systems.

4 Methodology

The requirements elicitation addresses five research questions (RQs) systematically derived from four sources: (i) D2.1 (Urban Event Mapping) providing event taxonomy, five-pathway interdependency framework, and empirical evidence; (ii) D2.3 (Establishing Urban States of Equilibrium) informing KPI definitions, modelling requirements, validation approaches, equity requirements, and governance frameworks; (iii) D2.5 competency question methodology linking ontological requirements to decision-support needs; and (iv) WP5 technical objectives defining SUMA's core functionality. These cross-references ensure RQs address both theoretical frameworks and operational pilot needs (Bratislava, Odesa, Larissa, AHEPA), guiding requirement extraction, validation, and traceability to WP5 outputs:

- **RQ1:** Who are the intended users and stakeholders of SUMA, and what decisions and workflows must the system support across planning, operations, and emergency contexts?

- **RQ2:** Which functional capabilities must SUMA provide to represent stressors, run scenario-based analyses, and assess resilience/antifragility in a way that is implementable within WP5?
- **RQ3:** What data, interoperability, and integration requirements are necessary for SUMA to operate with heterogeneous sources and connect to external simulation environments and tools?
- **RQ4:** Which non-functional, usability/UI, and governance/privacy requirements are essential to ensure SUMA is usable, trustworthy, secure, and compliant in real-world institutional settings?
- **RQ5:** How are requirements structured, linked to WP5 deliverables (D5.1–D5.4), and expressed with verification criteria to support implementation planning?

To address the above posited research questions, a staged workflow is proposed combining evidence synthesis, stakeholder engagement, and expert validation to ensure methodological robustness and practical relevance, as illustrated in Figure 1.

Requirements were systematically derived from different complementary sources, enabling triangulation between conceptual grounding and operational needs, as described in Section 4. First, Deliverable D2.1 (Urban Event Mapping) [33] established the analytical foundation by defining a structured event taxonomy, disruption classification logic, and impact characterisation framework. These elements informed requirements related to event modelling, cascading effects, and emergency response coordination. Second, a systematic literature review comprising 123 academic publications and technical reports provided the theoretical and methodological basis for antifragile urban mobility systems. The review covered domains including resilience engineering, smart mobility, digital twins, climate adaptation, governance coordination, and uncertainty-aware decision-support systems, thereby informing requirement formulation, indicator selection, and best-practice alignment.

Third, requirements were directly elicited from project partners and pilot cities (Bratislava, Thessaloniki, Larissa, and Odesa) through structured templates. This process captured priorities, contextual constraints, governance structures, and data availability conditions, ensuring that the emerging requirement set reflected real-world implementation environments rather than abstract system assumptions.

The integration of these sources produced an evidence-informed and de-duplicated requirement corpus that balances academic rigor with operational feasibility. Preliminary requirements were consolidated and organised into key thematic categories before undergoing refinement during a General Assembly workshop, as described in Section 6. This preparatory phase strengthened conceptual clarity and ensured analytical coherence prior to expert evaluation. To prioritise the requirement set under conditions of uncertainty, a Delphi-inspired expert consultation was subsequently employed as a structured validation stage, as reported Section 6.

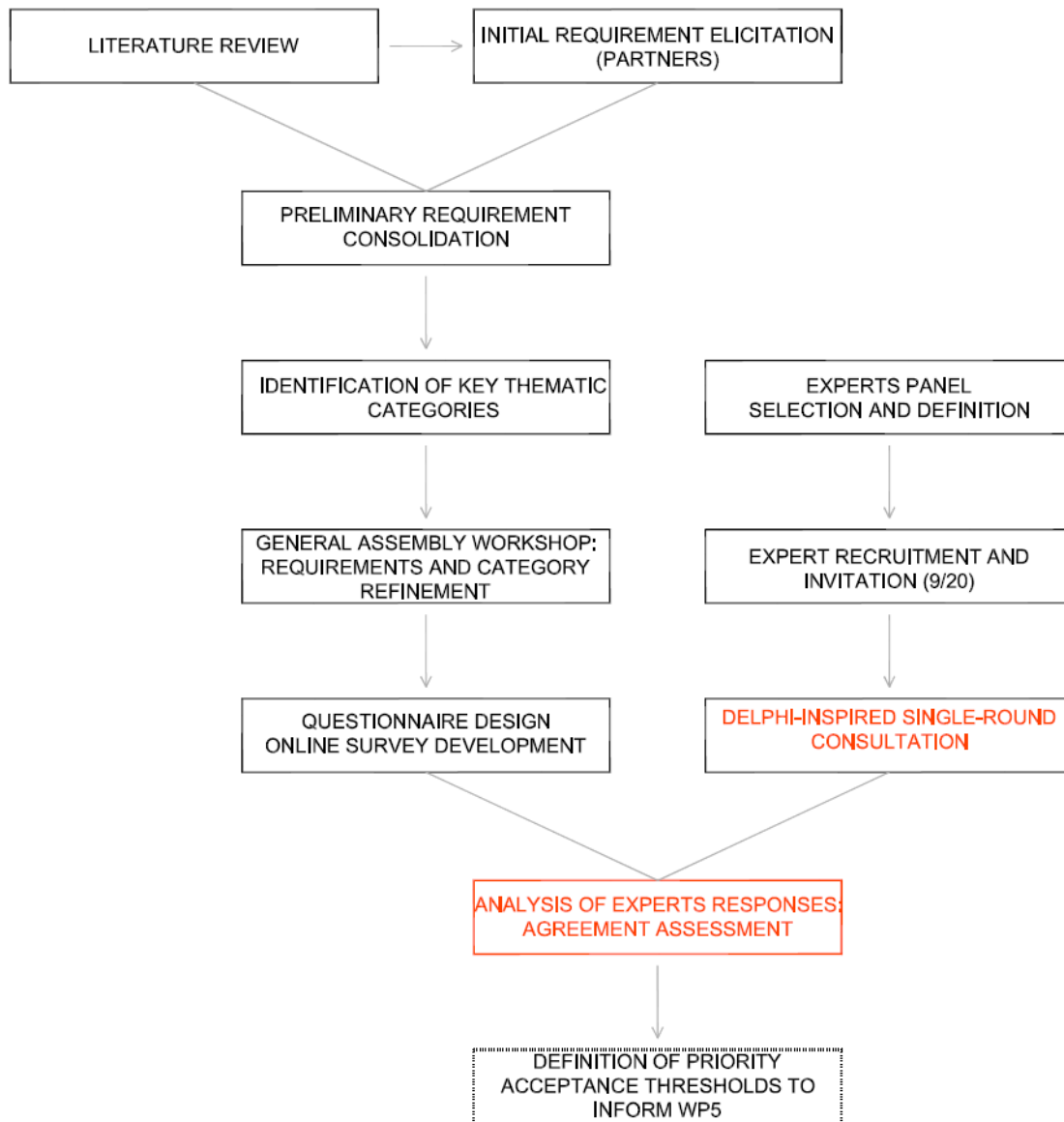


Figure 1 Methodology scheme.

By combining independent expert assessment with transparent agreement criteria, the consultation supported the identification of core requirements while systematically flagging items requiring further clarification. Delphi-inspired consultation in AntifragiCity is applied not only to rank traditional resilience requirements, but also to define and make actionable requirements linked to antifragility. In this context, antifragility refers to system behaviours that strengthen or improve when exposed to stress, disruption, or uncertainty. Through iterative expert input and structured feedback, the process helps identify which system features should adapt, learn, or evolve under stress, turning adverse conditions into opportunities for improvement rather than merely maintaining function.

In the following sections, background on the conceptual and ontology framework underpinning SUMA is introduced, defining the scope and structure used for requirement elicitation (Section 5). Section 6 reviews Delphi-based approaches to requirements elicitation and explains the rationale for the Delphi-inspired design adopted in this study. As peer-reviewed literature contains limited Delphi-based applications explicitly addressing antifragility in transport systems, with most studies still emphasising robustness, redundancy, and recovery, this gap motivates a structured expert elicitation approach to clarify, screen, and prioritise antifragility-oriented constructs (e.g., proactive risk management, real-time information sharing, inter-organisational collaboration, adaptive decision-making) in a transparent manner despite contextual variability and incomplete empirical evidence.

The results section then presents the structuring of the requirement set and the outcomes of the expert consultation. This is followed by the requirement specification framework and the detailed SUMA requirements catalogue. The final sections address requirements management and traceability, outline assumptions, constraints and open issues, and conclude with a discussion of the findings and their implications for WP5 implementation and iterative refinement. In the next section, these motivations are translated into a concrete stakeholder and user framing that anchors the requirement catalogue and its verification approach.

5 Conceptual foundation: ontology-driven requirements scope and Delphi-inspired elicitation

This section summarises the semantic foundation of our SUMA environment that the proposed requirements should address to provide the data structures and semantics to support our planned software functionality. This takes the form of an Ontology, developed and detailed as part of our Deliverable D2.5. As such for consistency purposes, the requirements elicitation exercise started as part of the ontology development process and were firmed up within the context of the present deliverable.

5.1 Ontology purpose and scope

The requirements are framed in the ontology framework developed in D2.5, which provides a structured conceptual foundation for representing urban mobility systems, disruptions, responses, and antifragility. This section defines the scope of SUMA requirements by presenting the ontology architecture that establishes conceptual boundaries and integration points for the system.

The scope of SUMA requirements is defined by the AntifragiCity ontology architecture (D2.5), which identifies the key conceptual dimensions necessary for modeling antifragile urban mobility systems. The ontology work in AntifragiCity follows the NeOn Methodology, a scenario-based framework for building ontology networks rather than a single monolithic ontology. The NeOn methodology is used as a process reference to structure ontology engineering activities and deliverable artefacts; it does not prescribe specific tooling or constrain data governance.

The NeOn methodology is used as a process reference to structure ontology engineering activities and deliverable artefacts; it does not prescribe specific tooling or constrain data governance. The ontology is delivered in World Wide Web Consortium (W3C)-standard RDF/OWL serialisations with stable identifiers, version metadata, and provenance annotations. Access conditions, licensing, and any restrictions on instance data publication are handled in accordance with the project data policy; where datasets cannot be openly shared, the ontology still provides reusable schema-level identifiers, mappings, and query templates without exposing restricted content. NeOn emphasises (i) collecting requirements and competency questions, (ii) reusing both non-ontological resources (documents, taxonomies, datasets) and existing ontologies, and (iii) supporting the evolution of a network of aligned ontologies over time.

In AntifragiCity, three NeOn scenarios are combined:

- **From requirements to implementation:** The Ontology Requirements Specification Document (ORSD) is defined by 61 requirements and their competency questions, derived from D2.1, the T1.2 governing variables, literature review and partner input.
- **Reusing non-ontological resources:** the core concepts for urban events, domains, scales, EM-DAT groups and social-media categories are mined from D2.1 and related project deliverables instead of invented from scratch.
- **Reusing ontological resources:** we reuse established ontologies and data models in a modular, pilot-driven manner. The core alignment set comprises: (i) built environment assets (IFC/BIM), (ii) spatial semantics (CityGML/INSPIRE and GeoSPARQL), (iii) transport operations and traffic events (Transmodel/NeTEx/GTFS/DATEX II), (iv) sensing and observations (SOSA/SSN, with SensorML/O&M used only where richer sensor metadata is required by pilot data), and (v) provenance and organisational metadata where needed (e.g., PROV-O, FOAF/SIOC). External

hazard vocabularies (e.g., Empathi, ENVO/SWEET) are used for event typing and alignment in the Urban Event Ontology module.

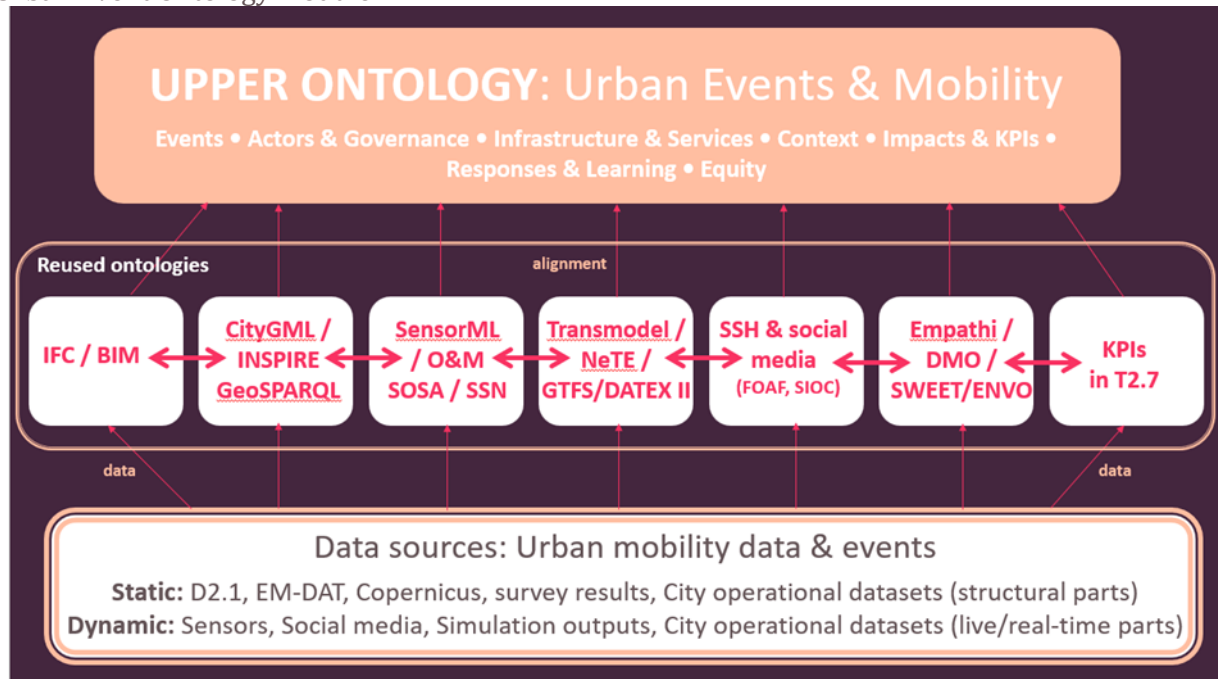


Figure 2. The architecture comprises three layers: upper ontology (master vocabulary), alignment layer (reused ontologies and standards D2.5), and data layer (event catalogues, operational datasets, sensor feeds).

Figure 2 illustrates the ontology architecture that defines the conceptual boundaries and integration points for SUMA requirements. This three-layer architecture organizes the ontology into distinct components following established ontology engineering principles [34].

Layers 1-2 define the T-BOX (Terminological Box), which constitutes the ontology schema consisting of classes, properties, and axioms that are shared across all pilot cities. Layer 3 populates the A-BOX (Assertional Box), which comprises the ontology instances consisting of individuals and facts that are city-specific.

- **Layer 1: Upper Ontology (Urban Events & Mobility)**

At the top sits the Upper Ontology: Urban Events & Mobility, which acts as the project's master vocabulary including a small set of high-level classes and relations that all other ontologies and datasets plug into. It captures the main facets required for antifragile urban mobility:

- **Events:** stressors and disruptions, classified by domain, scale, severity and type. The D2.1 event taxonomy is adopted as the reference taxonomy and implemented as a controlled vocabulary for event classification.
- **Actors & Governance:** individuals, organisations and authorities who operate and regulate the system.
- **Infrastructure & Services:** networks, facilities and mobility services that can be disrupted.
- **Context:** the wider background around an event, including environmental and PESTEL factors such as weather, economy, politics and social conditions.
- **Impacts & KPIs:** how disruptions affect mobility, people, economy and environment, and how this is measured (including antifragility scores).

- **Responses & Learning:** operational responses, long-term interventions and learning processes that improve the system after shocks.
- **Equity:** population groups, vulnerable users and fairness dimensions, reflecting the project's SSH objectives.

This upper ontology is intentionally lightweight but provides a consistent semantic layer for reasoning about events, their causes, contexts, impacts and responses across all pilot cities.

- **Layer 2: Alignment Layer (Reused Ontologies & Standards)**

The middle layer groups the reused ontologies that are aligned to the upper ontology:

- **IFC / BIM** for buildings and facilities at LOD 200-300 (schematic to detailed design level), so that stations, terminals and other key structures can be described using BIM semantics with sufficient spatial and semantic detail for mobility analysis, disruption assessment, and evacuation planning.
- **CityGML / INSPIRE / GeoSPARQL** for the urban spatial layer, including transport networks, land use, geometries and administrative units.
- **SensorML / O&M / SOSA / SSN** for sensors and observations e.g. traffic detectors, weather and air-quality sensors and other monitoring systems.
- **Transmodel / NeTEx / GTFS / DATEX II** for the operational transport layer: routes, stops, timetables and traffic events in public transport and on the road network.
- **SSH & social-media** vocabularies such as **FOAF** and **SIOC**, to represent users and organisations online, social-media posts and interaction metadata.
- **DMO (Disaster Management Ontology) / Empathi** ontology for structuring pre-disaster and post-disaster responsibilities and tasks and emergency-management ontology focused on representing emergencies, events, locations and impacts (used for social-media analysis and situation awareness). A dedicated Urban Event Ontology module is introduced in the middle layer to represent hazard and event types as first-class concepts. This module is derived from the D2.1 event taxonomy (formalised as a lightweight OWL vocabulary) and aligned where appropriate to external hazard/event resources (e.g., Empathi, ENVO/SWEET). This vocabulary is used to type event instances ingested from the bottom-layer catalogues/feeds (including the D2.1 Urban Event Catalogue). This makes event typing explicit and queryable, independent of whether event instances originate from historical catalogues or live feeds.
- A **KPI** ontology module integrating three complementary KPI sets: (i) D2.3 Core Antifragility KPIs (12 metrics including Antifragility Factor and System Responsiveness Index for equilibrium analysis), (ii) Task 2.7 Core KPI Subset (13 stakeholder-prioritised KPIs, selected by ≥ 4 stakeholders for cross-city comparability), and (iii) Extended KPI Catalogue (209 KPIs organized into 7 groups and 20 subgroups for city-specific analysis).

Vertical arrows labelled “alignment” represent mappings from concepts in these reused ontologies to the upper ontology. For instance, a DATEX II traffic event is aligned to the generic disruption-event class, a GTFS stop to a transport node, and Empathi/SWEET hazard classes to triggering-event types in the D2.1 event ontology.

- **Layer 3: Data Layer (A-BOX: Ontology Instance Population)**

The Data Layer represents the assertional component (A-BOX) of the ontology, populating the conceptual schema (T-BOX defined in Layers 1-2 and D2.5) with real-world instances. The bottom layer collects the data sources that instantiate ontology classes and properties: the D2.1 urban event catalogue, EM-DAT and Copernicus products, static city operational datasets (network topology, facilities, census and land-use layers), survey results, and dynamic sources such as sensor data, live operational feeds, simulation outputs and social-media streams. Static resources (D2.1, EM-DAT, Copernicus, survey snapshots) are distinguished from dynamic resources (sensors, social media, simulation outputs) based on ingestion and update patterns. Because the

middle-layer vocabularies are aligned to the upper ontology, all sources contribute to one integrated urban-mobility knowledge graph instead of remaining as disconnected silos.

Instantiation Model: While the ontology structure (T-BOX: classes, properties, relationships) is shared across all pilot cities, the A-BOX is instantiated separately for each city. This means there will be one ontology instance per pilot city (Bratislava, Thessaloniki, Larissa, Odesa), each containing:

- The same T-BOX (conceptual schema defined in D2.5 and refined through Delphi consultation)
- City-specific A-BOX (individuals and facts populated from that city's data sources)

For example:

- **Bratislava instance:** Contains individuals like Bratislava_Flood_March2024 (instance of DisruptionEvent class) with property hasRecoveryTime = "P7D"
- **Thessaloniki instance:** Contains individuals like Thessaloniki_Metro_Line1 (instance of MobilityService class) with property hasCapacity = "30000"

This approach enables city-specific reasoning and analysis while maintaining cross-city comparability through the shared conceptual framework.

Following ontology engineering best practices [34], [35] the three-layer architecture maintains a clear distinction between the T-BOX and A-BOX components of the ontology.

The T-BOX (Terminological Box) constitutes the ontology schema and spans Layers 1-2. It consists of classes, object properties, data properties, axioms, and constraints that define the conceptual structure of the ontology. This component is specified in D2.5 and has been validated and refined through the Delphi consultation process documented in this deliverable. The T-BOX is shared across all pilot cities, meaning the schema is city-agnostic and can be applied universally. For example, the T-BOX includes the class DisruptionEvent with subclasses such as NaturalHazard, InfrastructureFailure, and SocialDisruption. It also defines object properties like affectsService (with domain DisruptionEvent and range MobilityService) and data properties such as hasRecoveryTime (with domain DisruptionEvent and range xsd:duration). The T-BOX exhibits high stability, changing only through formal ontology versioning and governance processes.

In contrast, the A-BOX (Assertional Box) represents the ontology instances and corresponds to Layer 3. The A-BOX comprises individual entities including instances of the classes defined in the T-BOX, along with their property values and factual assertions. This component is populated in Layer 3 from pilot city data sources as described in the previous section. Unlike the T-BOX, the A-BOX is city-specific, with each pilot city maintaining its own distinct A-BOX. For instance, the Bratislava A-BOX might contain an individual Bratislava_Flood_March2024 as an instance of the NaturalHazard class, with property assertions such as Bratislava_Flood_March2024 affectsService Bratislava_Metro_Line1 and data values like Bratislava_Flood_March2024 hasRecoveryTime "P7D"^^xsd:duration (indicating a seven-day recovery period). The A-BOX is inherently dynamic, being updated continuously as new events occur, scenarios evolve, and monitoring data accumulates.

This T-BOX/A-BOX separation yields several practical implications for the project. First, it enables reusability: the T-BOX can be applied to cities beyond the four pilots without modification, facilitating future extensions of the framework. Second, it supports scalability, as new cities require only A-BOX population rather than T-BOX redefinition, significantly reducing implementation effort. Third, it simplifies maintenance by allowing schema evolution (T-BOX changes) to be governed separately from routine data updates (A-BOX changes), reducing the risk of unintended consequences. Fourth, it enables sophisticated reasoning capabilities: SPARQL queries and reasoning rules operate on the stable T-BOX structure while retrieving city-specific facts from the A-BOX,

allowing both general pattern detection and context-sensitive analysis. Finally, it ensures interoperability by allowing the shared T-BOX to enable cross-city comparisons, such as queries comparing recovery times for flood events across all cities, while preserving the local context captured in each city's A-BOX.

5.1.1 Domain-Independent Meta-Model: 12 Top-Level Classes

The ontology is organized around 12 domain-independent top-level classes that represent the fundamental aspects of antifragile urban mobility systems. These classes provide a stable semantic backbone that guided the requirements elicitation process by ensuring comprehensive coverage across all necessary dimensions (Figure 3).

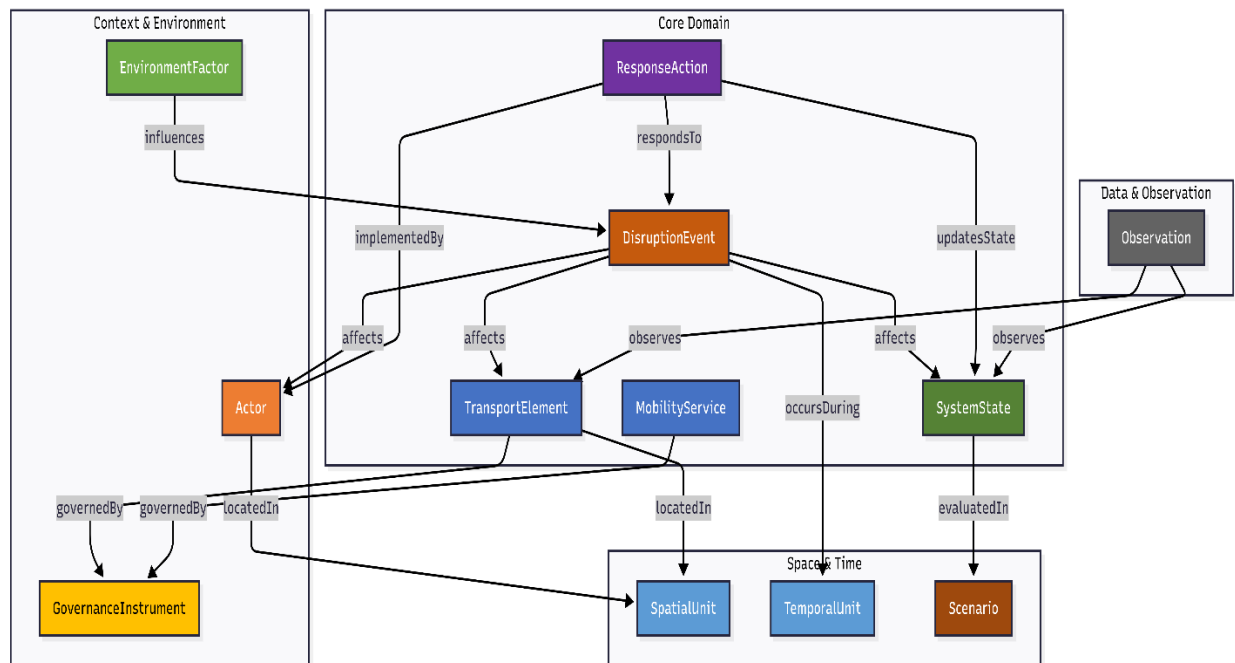


Figure 3. Meta-Model 12 Top-Level Classes (adapted from D2.5). This compact visual summary shows the upper-level backbone used to keep all modules consistent.

The 12 top-level classes are:

1. **TransportElement** Physical and digital infrastructure components (nodes, edges, facilities, vehicles, sensors, edge computing nodes) that constitute the mobility network.
2. **MobilityService** Service offerings and operations (public transport routes, shared mobility services, demand-responsive transport, accessibility services) that provide mobility to users.
3. **DisruptionEvent** Stressors, shocks, and cascading failures that disturb normal system operations, classified by domain, scale, severity and type according to the D2.1 event taxonomy.
4. **SystemState** Performance metrics, equilibrium states, and antifragility indicators (AntifragilityScore, ResilienceIndex, FragilityMetric, PerformanceDelta) that characterize system behavior over time.
5. **Scenario** Planning and simulation contexts (baseline scenarios, disruption scenarios, intervention scenarios, what-if analyses) used for decision-support and stress-testing.

6. **EnvironmentFactor** PESTEL context (Political, Economic, Social, Technological, Environmental, Legal) and climate factors (weather patterns, climate hazards, environmental conditions) that shape system vulnerability and adaptation capacity.

7. **Actor Stakeholders**, users, equity groups, organizations, and authorities who interact with, operate, or regulate the mobility system.

8. **GovernanceInstrument** Policies, regulations, coordination mechanisms, legal frameworks, and institutional structures that govern mobility system behaviour and responses.

9. **SpatialUnit** Geographic and network topology representations (administrative boundaries, corridors, zones, network graphs) that situate mobility elements in space.

10. **TemporalUnit** Time scales and temporal patterns (time-of-day, day-of-week, seasons, long-term trends) relevant to mobility dynamics and disruption evolution.

11. **Observation** Data, sensors, KPIs, measurements, data provenance, and trust metadata (DataSource, Provenance, TrustScore, Confidence, Timestamp) that support evidence-based decision-support.

12. **ResponseAction** Interventions, operational responses, policy measures, learning processes, and adaptive transformations that modify system behaviour to improve antifragility.

These 12 classes are domain-independent in the sense that they can be reused in different cities and possibly in other infrastructure domains and provide a stable schema even if lower-level specializations evolve. At the meta-model level, high-level object properties capture core relationships between these classes (e.g., affects, triggers, locatedIn, governedBy, observes, responds). These abstract properties are specialized for specific contexts (e.g., affectsEquity, affectsAccessibility, locatedInVulnerableArea, governedByRegulation) and support generic reasoning patterns such as impact chains and state transitions.

5.1.2 Multi-Source Requirements Derivation

Building on this ontological foundation, requirements were systematically derived from five complementary sources:

1. **D2.1 deliverable (Urban Event Mapping and taxonomy)**: Established the analytical foundation by defining a structured event taxonomy, disruption classification logic, and impact characterization framework.

2. **Systematic literature review (123 Q1 papers)**: Provided the theoretical and methodological basis for antifragile urban mobility systems across domains including resilience engineering, smart mobility, digital twins, climate adaptation, governance coordination, and uncertainty-aware decision-support.

3. **Ontology deliverable (D2.5)**: Provided 61 ontology requirements and competency questions, the 12-class conceptual meta-model, KPI catalogue (209 KPIs with 13 core stakeholder-prioritized KPIs), and integration framework aligning with established standards.

4. **Project partners and pilot cities (Bratislava, Thessaloniki, Larissa, Odesa)**: Direct elicitation through structured templates captured operational priorities, context-specific constraints, governance structures, data availability conditions, and pilot-specific resilience challenges.

5. **General Assembly workshop (December 2025, Bratislava)**: Structured expert workshop refined preliminary requirement formulations, validated analytical category structure, identified gaps and overlaps, and ensured alignment with pilot implementation contexts.

5.2 Ontology theoretical foundation

The requirements framework is grounded in the AntifragiCity ontology design principles (D2.5), which implement antifragility theory and ensure comprehensive coverage of socio-technical urban mobility systems. Critically, antifragility is treated as a first-class concern and central organizing principle throughout the ontology architecture, not as a secondary label or add-on consideration. Following Taleb (2012), the framework addresses systems that not only withstand shocks but demonstrably improve when exposed to instability—a fundamental shift from traditional resilience frameworks that focus solely on recovery to baseline performance. This antifragility-first philosophy shapes every aspect of the ontology design, and by extension, the SUMA requirements derived from it.

The ontology was designed according to the following principles that directly informed the requirements elicitation methodology:

5.2.1 Requirements- and Competency Question-driven

Ontology is directly grounded in the requirements and competency questions. Every core concept (e.g. `DisruptionEvent`, `SystemState`, `Scenario`, `ResponseAction`) appears in multiple requirements and is necessary to answer one or more CQs. Traceability is ensured by keeping `Requirement_ID` references at annotation level. This ensures that all SUMA requirements can be traced back to specific ontology concepts and validated through competency questions that define testable system capabilities.

5.2.2 Antifragility as a first-class concern

Antifragility is **not treated as a secondary label** but as a central organising principle. Following Taleb's conceptualization, the framework addresses systems that not only withstand shocks but improve when exposed to instability. This is reflected in the ontology design through:

- **SystemState** includes antifragility-specific indicators (`:AntifragilityScore`, `:ResilienceIndex`, `:FragilityMetric`, `:PerformanceTs`, `:PerformanceDelta`) that enable quantitative assessment of system improvement under stress.
- **DisruptionEvent** and **ResponseAction** are explicitly linked via patterns such as “disruption → response → updated system state”, enabling the system to track how interventions affect antifragility metrics over time.
- The ontology accommodates reasoning over **learning** and **structural change** (e.g. `:LongtermLearning`, `:StructuralChange`, `:AdaptiveTransformative` state), allowing SUMA to model not only immediate operational responses but also long-term institutional and structural adaptations that enhance antifragility.

These design choices ensure that SUMA requirements address both short-term resilience (bouncing back) and long-term antifragility (improving through adversity).

5.2.3 Socio-technical and cyber-physical integration

The ontology simultaneously represents **physical and digital infrastructure** (e.g. nodes, edges, sensors, edge nodes, analytics engines) as a **cyber-physical system**, and **actors, institutions, policies, equity groups and behavioural processes** as a **socio-technical system**. This dual representation acknowledges that urban mobility antifragility depends not only on technical robustness but also on institutional adaptability, social equity, and governance coordination.

This leads to **two strongly interconnected sets of modules**:

- Technical modules (Transport Network, Data & Analytics, System State, Disruptions), addressing the physical and digital infrastructure that enables mobility services and their monitoring.
- Socio-institutional modules (Actors & Equity, Governance & Policies, Response & Learning), addressing the human, organizational, and regulatory dimensions that shape mobility system behaviour and adaptation capacity.

Requirements derived from this framework therefore span both technical capabilities (e.g., real-time data integration, predictive analytics) and institutional capabilities (e.g., multi-stakeholder coordination, equity monitoring, policy-aligned interventions), ensuring SUMA supports holistic antifragility rather than purely technical resilience.

5.2.4 Minimal yet expressive top-level schema

To balance **expressivity** and **maintainability**, the ontology is organised around a **small set of top-level classes** (12), which are then specialised as needed. This simplifies alignment with external vocabularies, facilitates reasoning and querying, and reduces modelling effort when extending to new cities or use cases. By grounding requirements in this stable 12-class meta-model (Section 3.1, Figure 2), the framework ensures that SUMA requirements remain coherent and traceable across diverse pilot contexts while allowing flexibility for context-specific adaptations.

5.2.5 Open-world and incremental extension

The ontology follows the open-world assumption, which is typical of OWL ontologies: the absence of information does not imply falsity, and new concepts and individuals can be added without invalidating existing assertions, provided constraints are respected.

Modules are designed so that new pilots can extend existing classes with local subclasses, and new domains (e.g. energy, health) can connect via multi-layer network patterns.

This principle directly informs SUMA's requirements architecture: rather than specifying a closed, exhaustive set of requirements, the framework identifies a core baseline (Section 8) while supporting incremental extension as pilots identify additional context-specific needs during WP5 implementation. This aligns with the antifragility principle of adaptive evolution. The system's requirements can improve and expand through exposure to real-world deployment challenges.

5.2.6 Explicit uncertainty, provenance and constraints

Given the importance of predictive models, risk assessment, and legal constraints, uncertainty-related properties (:Uncertainty, :PredError, :ForecastUncertainty, qualitative scales) are explicitly modelled. Data provenance and trust (:DataSource, :Provenance, :TrustScore, :Confidence, :Timestamp, :Audit) have their own module, and legal and governance constraints (:LegalConstraint, :Regulation, :LegalScenario, :LegalBasis) are explicitly linked to actions and scenarios, enabling constraint checking.

These design choices ensure that SUMA requirements address not only functional capabilities but also the transparency, auditability, and trustworthiness necessary for real-world institutional deployment. Requirements related to data quality metadata (DR-02), governance and privacy (GR-01 through GR-04), and transparency of assumptions (GR-03) directly reflect these ontological principles, recognizing that antifragile systems require honest communication of uncertainty and robust provenance tracking to support informed decision-making under incomplete information.

The six ontological design principles described above (Sections 5.2.1–5.2.6) directly inform the SUMA requirements framework elicited through Delphi consultation (Sections 6–7) and validated against competency questions. Specifically, the theoretical foundation derived from the AntifragiCity ontology design principles (D2.5) ensures that SUMA requirements exhibit five critical characteristics, each traceable to specific ontological structures:

Traceability to competency questions: The requirements-driven ontology methodology (D2.5) establishes bidirectional links between every core ontology concept (e.g., DisruptionEvent, SystemState, Scenario, ResponseAction) and one or more competency questions that define testable system capabilities. This ensures SUMA requirements can be validated through concrete queries that the system must answer (e.g., "What backup power capacity is required at critical mobility nodes?" maps to requirements FR-03, DR-01, and the Infrastructure and ResponseAction ontology classes).

Antifragility as measurable system property: Unlike traditional resilience frameworks that focus on recovery to baseline, the ontology implements antifragility through explicit quantitative indicators in the SystemState class (D2.5), including :AntifragilityScore, :ResilienceIndex, :FragilityMetric, :PerformanceTs, :PerformanceDelta. These enable SUMA to assess whether interventions create durable capability gains beyond pre-shock baselines (transformative resilience with antifragile potential, as defined in D2.1), translating Taleb's theoretical concept into computable requirements (FR-02, FR-05, NF-01).

Technical and socio-institutional integration: The Layer 2 Domain Ontology (D2.5) comprises dual module sets: Technical modules (Transport Network, Data & Analytics, System State, Disruptions) and Socio-institutional modules (Actors & Equity, Governance & Policies, Response & Learning). This dual structure ensures requirements span both technical capabilities (e.g., FR-01 real-time data integration, FR-04 predictive analytics) and institutional capabilities (e.g., GR-01 multi-stakeholder coordination, GR-02 equity monitoring, GR-04 policy-aligned interventions), addressing urban mobility antifragility as a cyber-physical and socio-technical system rather than purely infrastructure resilience.

Stable conceptual backbone with incremental adaptation: The 12 top-level class meta-model and three-layer T-BOX/A-BOX architecture (D2.5) provide structural coherence: the ontology schema (T-BOX) is shared across all pilot cities, while city-specific instances (A-BOX) can be extended incrementally without invalidating core requirements. This design informs the requirements architecture: a core baseline (Section 8 Priority 1 requirements) is specified, while supporting incremental extension as pilots identify context-specific needs during WP5 implementation (Priority 2–3 requirements), aligning with the antifragility principle of adaptive evolution through real-world deployment.

Explicit uncertainty, provenance, and governance: The ontology explicitly models uncertainty (:Uncertainty, :PredError, :ForecastUncertainty), data provenance (:DataSource, :Provenance, :TrustScore, :Confidence, :Timestamp, :Audit), and legal constraints (:LegalConstraint, :Regulation, :LegalBasis) through dedicated modules (D2.5). This ensures requirements address not only functional capabilities but also transparency, auditability, and trustworthiness necessary for institutional deployment: DR-02 (data quality metadata), GR-01 through GR-04 (governance and privacy), GR-03 (transparency of assumptions), recognizing that antifragile systems require honest communication of uncertainty and robust provenance tracking to support informed decision-making under incomplete information.

These five characteristics ensure systematic alignment between ontological design (D2.5), competency questions, and Delphi-validated requirements (Sections 6–7), providing the conceptual foundation for SUMA's implementation in WP5.

5.3 Delphi and Delphi-inspired approaches to requirements elicitation

A comprehensive review of Delphi consultation literature is provided in the Appendix A, where detailed discussions of Delphi method fundamentals, requirements identification, methodological considerations, practical implementation, and applications in urban and mobility systems are presented.

Building on this literature, the deliverable positions the Delphi-inspired expert consultation as the central validation and prioritisation stage within a broader requirements elicitation framework. Importantly, the requirement set evaluated by experts was not generated through open-ended exploration but derived from multiple prior sources, including ontology development, systematic literature synthesis, project documentation, and stakeholder input. This prior structuring fundamentally shapes the methodological role of expert consultation, shifting its purpose from requirement discovery toward requirement validation, clarification, and prioritisation [36], [37], [38], [39], [40]. Such positioning aligns with modified and context-specific Delphi applications reported in applied socio-technical system research, where expert consultation is used to stabilise analytically grounded requirement sets rather than to produce conceptual convergence through repeated rounds of generation [36], [37], [38], [41]. In this configuration, Delphi principles, independence of judgement, structured assessment, and transparent agreement measurement, remain central, while iterative refinement may occur outside the formal Delphi rounds through subsequent implementation cycles and stakeholder engagement processes [21,22].

The Delphi method constitutes a structured expert elicitation approach widely employed to support decision-making in domains characterised by complexity, uncertainty, and limited empirical evidence [37], [42]. The method derives its name from the Oracle of Delphi, symbolizing its purpose of obtaining reliable consensus through structured expert consultation [38], [43]. Originally developed to obtain reliable judgement in forecasting contexts, Delphi relies on iterative consultation with a panel of experts combined with controlled feedback and statistical aggregation of responses [39], [42]. The methodological rationale underpinning Delphi is that collective expert judgement, when elicited under conditions of independence and structured reflection, can provide robust guidance in situations where experimental validation or historical data are insufficient [44], [45]. These characteristics explain the method's continued relevance across socio-technical research domains, including infrastructure planning, policy design, technology assessment, and complex system engineering [36], [43].

Within requirements engineering, Delphi approaches are commonly used to support the validation and prioritisation of requirement statements when multiple stakeholder perspectives must be reconciled [37], [46]. Requirements for complex systems frequently span technical, organisational, and governance dimensions, making them difficult to derive solely through analytical modelling or empirical observation [47], [48]. In such contexts, Delphi offers a mechanism to formalise tacit knowledge, clarify requirement interpretation, and identify areas of agreement and divergence among experts [37], [41]. Rather than functioning exclusively as a requirement generation tool, Delphi is frequently applied after an initial evidence-based structuring phase, enabling systematic assessment of requirement relevance, clarity, and implementation priority [46], [49].

Existing applications of Delphi in urban and mobility research illustrate this embedded role. Prior studies have used Delphi techniques to validate resilience indicators, prioritise policy measures, assess smart city performance criteria, and evaluate transport innovation pathways [50], [51], [52], [53]. These applications demonstrate the suitability of structured expert elicitation for capturing interdisciplinary knowledge and addressing uncertainty in planning contexts [36], [37], [43], [44], [45]. However, most Delphi-based studies in mobility research focus on resilience, robustness, or policy prioritisation rather than on the operationalisation of antifragility as a system design principle [53], [54]. As a result, there remains relatively limited methodological guidance on how expert elicitation can support the translation of antifragility theory into implementable software and decision-support requirements [3], [36], [37], [45], [53].

This gap is particularly relevant for decision-support environments intended to represent adaptive and learning-oriented system behaviour. Antifragility introduces requirements that are inherently dynamic, context-dependent, and difficult to specify through conventional requirement elicitation approaches alone [3], [44], [55], [56]. Expert judgement therefore becomes essential for clarifying how such concepts should be operationalised, prioritised, and interpreted within real-world planning and operational settings [36], [37], [43], [44], [45]. Delphi-based validation offers a structured mechanism for addressing this challenge by enabling systematic evaluation of requirement statements that capture adaptive capacity, learning processes, cross-system interactions, and governance coordination [36], [45], [53].

Accordingly, the methodological choice to adopt a Delphi-inspired design should not be interpreted as an attempt to position the approach as superior to classical multi-round Delphi processes. Instead, it reflects methodological alignment with the study's objectives, starting conditions, and implementation context. Classical Delphi designs are particularly appropriate when research seeks to generate new constructs, explore divergent viewpoints, or achieve progressive convergence across multiple rounds [43], [44]. By contrast, when requirement statements are already analytically grounded and the research objective concerns validation and prioritisation for implementation, Delphi-inspired approaches represent a methodologically established adaptation [36], [37], [38], [41]. In this sense, the Delphi-inspired consultation adopted in this study should be understood as an appropriate methodological configuration that supports the translation of antifragility concepts into implementable system requirements. In contexts where requirement statements are already analytically grounded and the research objective concerns validation and prioritisation for implementation, Delphi-inspired approaches provide a methodologically appropriate and widely recognised adaptation [41], [48].

5.3.1 Why Delphi-inspired in Antifragility setting

In AntifragiCity, a Delphi-inspired approach was adopted because the requirement set had already been substantially structured prior to expert consultation, as described in Section 4-5. As outlined in the preceding sections, requirements were derived from ontology development, systematic literature review, project documentation analysis, partner input, and refinement during the General Assembly workshop. Under these conditions, expert consultation was not intended to support requirement generation but to assess relevance, clarify interpretation, and establish priorities within an existing requirement set. The Delphi-inspired assessment therefore functions as a validation step that confirms consistency with the ontology, reflects stakeholder input collected through structured templates, and enables examination of requirements against competency questions used to support verification. This configuration is consistent with modified Delphi applications reported in applied socio-technical research, where expert consultation is used to

stabilise analytically grounded requirement sets rather than to support construct discovery through repeated rounds [36], [37], [40]. The methodological choice reflects the starting conditions of the study. Classical multi-round Delphi designs are appropriate when conceptual constructs remain open or divergent viewpoints require progressive convergence. By contrast, when requirements have been consolidated in advance, structured expert elicitation primarily supports clarification, prioritisation, and implementation readiness [36], [40].

The decision to employ a single prioritisation round also affects the interpretation of agreement. Round-to-round stability measures commonly used in classical Delphi studies are not applicable. Agreement is therefore reported using within-round dispersion indicators (e.g. interquartile range), enabling transparent prioritisation while identifying lower-agreement items as candidates for subsequent refinement during WP5 implementation cycles.

More broadly, methodological adaptations of Delphi are common in applied requirements engineering, where the primary objective is the formalisation of implementable requirement baselines [36]. Prior research in complex urban cyber-physical systems indicates that requirement robustness depends on combining structured expert judgement with complementary evidence sources and maintaining explicit links between requirement intent, verification, and traceability [57]. The Delphi-inspired design adopted here follows this logic by positioning expert consultation as a validation mechanism within a mixed-method requirements pipeline rather than as a standalone consensus-generation procedure [36].

5.3.2 Delphi- inspired design for requirements prioritisation

This study employed a structured expert elicitation approach based on a Delphi - inspired design consisting of a single prioritisation round. While the project Description of Action (DoA) outlined a three-round Delphi process, this modified single-round design was adopted, on the basis that the requirement catalogue had already been substantially pre-structured through ontology derivation, literature synthesis, and the General Assembly workshop, thereby fulfilling the iterative refinement function that additional rounds would otherwise serve. The modified Delphi approach was preferred for three primary reasons. First, the requirement catalogue had already been systematically structured through literature review and stakeholder engagement, making prioritisation more appropriate than exploratory concept generation. Second, the method aligns with the project's applied objectives by supporting the validation of implementable requirements rather than pursuing purely theoretical consensus. Third, the approach offers an analytically robust yet operationally efficient framework for expert elicitation, preserving methodological rigor while facilitating timely decision support.

The design retains the core principles of Delphi research, independent expert judgment, structured data collection, and transparent agreement criteria, while adapting them to the needs of a complex socio-technical decision-support context. The consultation was preceded by targeted literature analysis and partner-based requirement elicitation to establish an evidence-informed foundation and reduce interpretative ambiguity. These preparatory activities are consistent with modified Delphi protocols, where early structuring enhances the reliability of subsequent expert evaluation. The resulting inputs were consolidated into a harmonised and de-duplicated requirement catalogue expressed in consistent terminology to support unambiguous assessment and later verification. An expert panel was then defined, recognising that the validity of Delphi-type studies depends on the expertise and relevance of participants. Twenty experts were invited through a formal selection process, resulting in nine confirmed participants and ensuring transparency in panel formation. Independent response collection further reduced the risk of dominance effects and

supported balanced evaluation. A preparatory workshop (General Assembly) contributed to refining the requirement set and validating its thematic organisation, informing the development of a structured questionnaire. The consultation was subsequently implemented through an online survey to enable standardised data collection and systematic analysis. Expert responses were analysed to identify and rank priority requirements based on predefined agreement criteria. Consensus-oriented studies commonly establish thresholds in advance, with agreement levels around 70–80% frequently reported in Delphi research.

5.3.3 Extrapolation of requirements for Delphi-inspired expert consultation

The requirements included in the Delphi-inspired single-round expert consultation were derived through a structured and evidence-informed process designed to ensure traceability, methodological rigor, and relevance to the AntifragiCity pilots and prototypes. As illustrated in Figure 1, in Section 4, the workflow comprised the following sequential steps: i) literature review, ii) initial requirement elicitation involving project partners, iii) preliminary requirement consolidation, and identification of key thematic categories. These preparatory activities preceded the expert consultation and ensured that the requirement set was analytically grounded and suitable for structured evaluation. The initial 61 requirement pool resulted from a comprehensive analysis of AntifragiCity project materials (including objectives, pilot descriptions, and expected outputs) combined with a targeted review of the scientific and technical literature on antifragile urban mobility systems. Rather than extracting requirements mechanically, the analysis was guided by a set of research questions that functioned as analytical lenses to identify both explicit and implicit requirements across project documentation and scholarly sources, as described in Section 4.

The research questions were formulated to span the full system lifecycle and capture technical, operational, governance, and societal dimensions relevant to pilot implementation. Each question supported the systematic screening of the literature and project materials, enabling the extraction of statements reflecting necessary conditions, functional capabilities, or design constraints for antifragile urban mobility systems.

Extracted statements were reformulated into candidate requirements and subsequently organised into key thematic categories to facilitate structured analysis. During this process, semantically overlapping or redundant statements were consolidated, while context-specific implementations were abstracted into more general, function-oriented requirements. This abstraction ensured that the resulting requirement set remained transferable across heterogeneous urban contexts while preserving traceability to the underlying evidence base. **Table 1** summarises the analytical domains that guided the extraction process.

Table 1. Analytical domains used to guide this extraction process.

Domain	Analytical focus
Technical and data infrastructure	Data sources, interoperability, latency, privacy, security
Modelling and analytics	Simulation, prediction, optimization, uncertainty
Decision-Support and governance	Triage mechanisms, institutional roles
Citizen and Stakeholder Engagement	Living Labs, co-creation, feedback integration
Evaluation and learning	KPIs, impact assessment, system learning
Scalability and sustainability	Transferability, adaptability, long-term operation

Following extraction, candidate requirements were consolidated to a higher level of generality to support expert judgment. Highly granular or implementation-specific statements were merged into broader requirement characteristics, reducing cognitive load while maintaining analytical coherence.

At this stage, no expert assessment or agreement measurement was conducted. Instead, recurrence across sources was used solely as an internal structuring mechanism to distinguish widely reported requirements from less frequently discussed or emerging aspects. This step informed the organization of the requirement set without constituting a validation or prioritization exercise. Recurrence patterns were used to organise candidate requirements and to distinguish between core, context-dependent, and emerging aspects, as summarised in Table 2.

Table 2 Recurrence patterns used to organise candidate requirements

Recurrence pattern	Interpretation
Frequently reported across sources	Core requirement to be evaluated by experts
Moderately reported	Context-dependent requirement
Rarely reported	Emerging or exploratory requirement
Divergent descriptions	Topic requiring expert deliberation

The consolidated requirements were subsequently translated into neutral, non-prescriptive statements suitable for questionnaire development. Wording was intentionally kept solution-agnostic, enabling experts to independently evaluate the importance and relevance of each requirement irrespective of specific technologies, urban settings, or institutional arrangements.

The structured requirement set was later subjected to refinement during a General Assembly (GA) workshop prior to questionnaire finalisation and administration within the Delphi-inspired single-round consultation.

5.3.4 Requirements for Delphi-inspired consultation

Table 3 synthesises the 47 requirements developed for the Delphi-inspired expert consultation. The candidate requirements, derived from the evidence extraction and consolidation process, were structured into a coherent set of functional and strategic capabilities to support systematic expert evaluation. Each requirement is formulated as a neutral, solution-agnostic statement to ensure consistency in expert judgment regarding relative importance and prioritisation.

Table 3 represents the 47 candidate requirements (DEL-R1 to DEL-R47) developed through the structured evidence extraction process. These solution-agnostic statements capture functional and strategic capabilities across technical, governance, equity, and antifragility dimensions, enabling systematic prioritisation by experts.

Table 3 Requirements

No.	Title	Description
DEL-R1	Cybersecurity and Privacy Protection	Protecting against cyber threats and ensuring user data privacy through security measures, privacy-preserving techniques, incident response protocols, and operational continuity.
DEL-R2	PESTEL Contextual Integration	Integrating Political, Economic, Social, Technological, Environmental, and Legal contextual factors into decision-support processes with transparent trade-off presentation.
DEL-R3	Evacuation and Rerouting Prioritisation	Pre-defined evacuation priority frameworks with traffic management capabilities, phased timing, and provisions for individuals with special mobility needs.
DEL-R4	Cross-City Knowledge Transfer	Sharing and comparing resilience strategies between cities for peer learning, benchmarking, and dissemination of validated best practices.
DEL-R5	Urban Planning for Resilient Densification	Strategic urban planning interventions to create resilient, accessible urban forms through densification of high-demand areas, development of peripheral sub-centres, and integration of mobility and land-use policy aligned with 15-minute city principles.
DEL-R6	Integration of Public Transport, Pedestrian and Bicycle Data	Ensuring data coverage includes all transport modes with equal visibility and analytical treatment for balanced assessment of network-wide conditions.
DEL-R7	Quick Response to Events	Real-time data ingestion from multiple sources with latency not exceeding 60 seconds, and rapid automated or one-click responses deployable within 2 minutes of event detection.
DEL-R8	Stakeholder Coordination	Supporting coordinated decision-making across multiple stakeholders through shared platforms, common operating pictures, and pre-defined coordination protocols.
DEL-R9	Environment-Friendly Traffic Control	Traffic management approaches that deliver environmental co-benefits alongside resilience, incorporating antifragile control algorithms that reduce vehicle idling time and associated pollutant emissions.
DEL-R10	Centralised Event Management	Tools for unified command and control during major events, with automatic or one-click activation of flow redirection protocols and cross-agency coordination.
DEL-R11	Real Traffic Data	Integration of live traffic data from multiple sources covering all major road segments, updated at intervals not exceeding 5 minutes with anomaly detection for unusual conditions.

DEL-R12	User-Friendly and Versatile Functions	Intuitive interfaces delivering accurate messages across multiple channels, with multilingual and accessible content tailored to different audiences.
DEL-R13	Interdisciplinary Propagation of Disruptive Events	Recording and tracking how disruptive events propagate across different urban system components, capturing cascading effects and cross-sector failure chains.
DEL-R14	Interactions Between Transport Modes	Managing interdependencies and coordination between different transport modes to prevent cascading failures, with real-time data sharing and integrated control across all modes.
DEL-R15	AI-Based Predictive Maintenance	Using AI/ML models to predict when infrastructure components are likely to fail, enabling proactive scheduling of repairs before disruption occurs.
DEL-R16	Multi-Level Coordination and Synchronisation	Coordination between different levels of government through shared dashboards and policy synchronisation to prevent conflicting actions across jurisdictions.
DEL-R17	Real-Time Monitoring and Analytics	Continuous 24/7 monitoring of mobility conditions with analytical insights, anomaly detection alerts, and comparison against historical baselines.
DEL-R18	Citizen Mobility Condition	Accounting for citizens' specific mobility constraints to tailor routing recommendations and provide priority evacuation support during emergencies.
DEL-R19	Behavioural Nudging for Sustainability	Using user preferences and past disruption responses to nudge travellers toward more sustainable choices such as public transport, cycling, off-peak travel, and carpooling.
DEL-R20	Severity and Impact Scales	Context-specific qualitative severity and impact scales calibrated through stakeholder input and consistently applied across all system outputs.
DEL-R21	Multi-Modal Data Fusion	Combining data from multiple transport modes and sources into a unified analytical framework, identifying cross-modal correlations and reconciling discrepancies between sources.
DEL-R22	Data Provenance and Trust Management	Tracking data sources and managing trust through provenance recording, source trust scoring, and cross-validation protocols for critical data types.
DEL-R23	Resilient Edge Processing Units	Distributed computing resources at the network edge capable of maintaining essential control functions during communication outages or central system failures.

DEL-R24	Economic Impact Modelling	Quantifying economic costs of disruptions, optimising resource allocation for recovery, and analysing distributional economic impacts across communities and sectors.
DEL-R25	Legislation Input	Incorporating relevant laws, regulations, and policy constraints into decision logic, with updatable regulatory repositories and support for emergency orders.
DEL-R26	Antifragility Score Calculation	Calculating metrics that quantify how much the city improves after disruptions, with temporal tracking and comparative benchmarking between cities or subsystems.
DEL-R27	Long-Term Transformative Learning	Continuous long-term institutional learning and adaptation, flagging when transformative changes may be warranted, and tracking performance trajectories over years and decades.
DEL-R28	Mobility of Emergency Response Services	Ensuring emergency vehicles can move unhindered during disruptions through signal pre-emption, optimised routing, and contingency plans for degraded network conditions.
DEL-R29	Possible Scenario Evaluation	Real-time evaluation of alternative response options with comparative outcome projections completed within 5 minutes for time-critical decisions.
DEL-R30	Usability Across Split Governance	Role-based access control and data partitioning enabling different entities to access appropriate data and functions within a defined trust framework.
DEL-R31	Antifragile Against Disruptions	The system learns from smaller disruptions to improve performance under larger ones, demonstrating improved throughput, delay reduction, or recovery time as disruption intensity increases.
DEL-R32	Prioritisation of Emergencies	Dynamic incident prioritisation based on severity, escalation potential, and impact on vulnerable populations, with standardised severity classification.
DEL-R33	Graph Database	Using graph databases to model the urban mobility network, representing nodes and edges with their attributes to support efficient pathfinding, disruption impact analysis, and real-time topology updates.
DEL-R34	Stakeholder Input	Enabling authorised stakeholders to update the system's database with relevant information through secure input mechanisms, with audit trails for accountability.
DEL-R35	Geographical Limitations Input	Incorporation of geographic and environmental constraints—including terrain, protected areas, flood zones, and climate-related hazards—into all routing, planning, and emergency response algorithms.

DEL-R36	Learn to Reroute Proactively	Predictive detection of emerging disruptions and proactive traffic rerouting before congestion or blockage fully develops, with continuous improvement of predictive accuracy.
DEL-R37	Equitable Access and Inclusive Mobility	Ensuring fair access for all population groups including underserved communities, meeting accessibility standards, and preventing disproportionate disadvantage during disruptions.
DEL-R38	KPI-Driven Performance Monitoring	Monitoring a comprehensive suite of Key Performance Indicators with real-time dashboards, periodic reports, threshold alerts, and policy-KPI linkage.
DEL-R39	Real-Time Equity Monitoring	Monitoring how mobility conditions are distributed across geographic areas and demographic groups, with equity dashboards and alerts for disproportionate outcomes.
DEL-R40	Active Mobility	Emphasising walking and cycling in the mobility mix with supporting infrastructure, system integration, and prioritisation during disruptions to motorised transport.
DEL-R41	Simulation of Unexpected Events	Capability to simulate hypothetical disruption scenarios for stress testing, vulnerability identification, operator training, and preparedness exercises.
DEL-R42	Flexible Route and Fleet Planning	Dynamic adjustment of vehicle routes and fleet deployments in response to real-time conditions, enabling rapid redeployment without requiring pre-scheduled service changes.
DEL-R43	Evaluating Transport Measure Impacts	Ex-ante and ex-post assessment of transport policies and projects on resilience KPIs, with results feeding into policy feedback loops.
DEL-R44	Maximise Individual Mobility	Optimising mode allocation to move the most people to their destinations as rapidly as possible, dynamically assigning transport modes based on real-time demand patterns rather than optimising for vehicle throughput alone.
DEL-R45	User Choice	Incorporating user preferences and providing viable alternatives rather than single solutions, learning from behaviour while respecting privacy and user autonomy.
DEL-R46	Energy Optimisation and Emission Reduction Control	Adaptive signal timing and control strategies to minimise energy consumption and emissions by reducing average vehicle idling time during peak hours.
DEL-R47	Scenario Analysis and Forecasting System	Decision-support tools for projecting likely outcomes of disruption scenarios or intervention options, including policy impact and evolving crisis forecasting.

6 Results of Delphi-inspired consultation

This section presents the results of the requirement structuring and expert consultation process, including the refinement of analytical categories and the composition of the Delphi panel, which together establish the foundation for the subsequent prioritisation analysis.

6.1 Categories

The initial categorisation was derived from a structured literature review and reflected the core dimensions of urban resilience most consistently identified in established academic frameworks. As presented in Figure 1, this initial set of categories supported the identification of key thematic areas prior to expert consultation and was therefore conceived as a provisional analytical structure rather than a definitive taxonomy.

A General Assembly (GA), held in Bratislava on 10-11 December 2025, functioned as a structured expert workshop within the preparatory phase of the Delphi-inspired single-round consultation. During this session, multidisciplinary project partners critically assessed the adequacy, scope, and granularity of the preliminary categories. This collective review highlighted the need to further disaggregate several cross-cutting dimensions that were insufficiently explicit in the original framework.

As a result, the category of risk governance was refined to explicitly incorporate capacity-building considerations, while three additional categories, safety and security, policymaking, and data and feedback management, were introduced to enhance conceptual clarity and operational relevance. This targeted refinement strengthened the analytical resolution of the framework while maintaining alignment with the evidence-informed foundation established during the literature review and requirement consolidation stages.

Importantly, this step represents a structured refinement activity within the consultation preparation process rather than a consensus-building exercise. The redefined categories provided the conceptual basis for questionnaire development and subsequent expert evaluation, ensuring that the requirement set was organized according to dimensions considered both theoretically grounded and operationally meaningful.

Table 4 presents the category redefinitions following the General Assembly.

Table 4 Refinement of analytical categories after the general assembly review

No. categories	Description from literature review	Categories optimisation after GA
1	Urban morphology & land use	Urban morphology & land use
2	Environmental resilience & climate adaptation	Environmental resilience & climate adaptation
3	Socio-technical infrastructure & mobility systems	Socio-technical infrastructure & mobility systems
4	Risk governance & emergency management	Risk governance & emergency management/capacity building
5	Social equity & community resilience	Social equity & community resilience
6	-	Safety and security
7	-	Policy making
8	-	Data and feedback management

6.1.1 Role of the GA in category refinement

The General Assembly convened internal project partners representing a broad range of disciplinary, institutional, and geographical perspectives. Within a Delphi-inspired methodological framework, this stage functioned as a structured pre-consultation refinement step undertaken prior to formal expert elicitation, with the objective of strengthening the conceptual robustness of the analytical framework rather than achieving consensus.

During this phase, participants were invited to review the formulation and completeness of the requirements, assessing whether they were comprehensive, clearly defined, and analytically sound. They subsequently evaluated the alignment between individual requirements and their assigned categories, with the aim of identifying potential misclassifications, redundancies, or unaddressed dimensions. Building on this review, participants examined each category to determine whether it: (i) adequately captured the requirements mapped to it; (ii) exhibited clear conceptual boundaries relative to other categories; and (iii) represented decision-relevant dimensions for pilot design and evaluation.

Feedback was collected through moderated discussion supported by iterative clarification, enabling the systematic identification of ambiguities, overlaps, structural gaps, and formulation inconsistencies within the preliminary categorisation. This structured review provided a robust basis for refining both the category definitions and the allocation of requirements, thereby improving the internal coherence and analytical resolution of the framework prior to the Delphi-inspired single-round consultation.

Three principal types of refinements emerged from the General Assembly review process. First, the scope and terminology of several categories were refined to better reflect their conceptual intent. Notably, the category initially labelled urban morphology was expanded to explicitly incorporate land-use considerations, acknowledging the interdependence between spatial form, land-use planning, and the performance of mobility and infrastructure systems. This adjustment improved alignment between category definitions and mapped requirements. Second, a distinct category on data and feedback management was introduced following the recognition that data governance, provenance, trust, and feedback mechanisms constitute a critical cross-cutting dimension insufficiently represented in the preliminary framework. Its inclusion addressed a structural gap and enhanced traceability for data-centric requirements. Third, governance-related requirements were reorganised to distinguish risk governance and emergency management from policymaking. This separation allows short-term operational decision-making and long-term strategic and regulatory processes to be treated as analytically distinct dimensions, thereby reducing conceptual overlap and improving interpretability.

6.2 Delphi panel design and expert selection

The construction of the expert panel followed established methodological guidance for Delphi-based research. As highlighted by Adler et al [49] experts participating in structured consultations should: (a) demonstrate willingness to participate, (b) possess domain-relevant expertise, (c) be able to allocate sufficient time to provide informed feedback, and (d) communicate their judgments in a clear and reasoned manner. Given the high level of domain expertise represented within the panel, analytical validity was prioritised over numerical panel size. These criteria guided both the identification and selection of panel members, ensuring alignment with the preparatory stages, namely the expert panel selection and definition followed by the recruitment and invitation process. The composition of expert panels is inherently research-dependent, and no universally accepted

panel size exists. Variability in panel size is commonly attributed to differences in research objectives, methodological design, and practical constraints such as time and expert availability [36], [42], [58]. Witkin et al. [59] suggest that panels typically remain below 50 participants, while Clayton [60] distinguishes between homogeneous panels, comprising experts from a single domain, and heterogeneous panels that integrate multiple areas of expertise. The latter configuration is particularly appropriate for complex and interdisciplinary topics such as urban resilience and mobility systems, where technical, governance, social, and infrastructural dimensions intersect. Empirical studies further illustrate this variability. For example, Alshehri et al. [52] initially engaged 71 experts, with 40 completing all three rounds, whereas Labaka [61] in research on critical infrastructure resilience, reported 15 complete responses from an initial panel of 21 experts. Such evidence underscores that methodological adequacy is determined less by absolute panel size than by the relevance, diversity, and engagement of participating experts. The composition of expert panels in Delphi-type studies is context-dependent, with analytical validity primarily determined by the domain expertise of participants rather than numerical panel size. Accordingly, twenty experts were invited to ensure disciplinary relevance and perspective diversity while maintaining procedural feasibility. Ultimately, nine experts confirmed participation, forming a panel consistent with recommended ranges for targeted expert elicitation. This recruitment approach aimed to secure a sufficiently diverse and knowledgeable group capable of providing informed evaluations while maintaining feasibility within the study's timeframe. Ultimately, nine experts confirmed participation, establishing a panel size consistent with methodological guidance for structured expert elicitation and appropriate for the focused scope of the consultation. As shown in Figure 4, the geographic distribution of the expert panel reflects a diverse international representation, encompassing participants from eight countries across Europe, North America, and Oceania. Greece accounts for the highest representation with two experts, while Australia, Denmark, Luxembourg, the United Kingdom, the Netherlands, the United States, and Italy each contribute one expert. This distribution demonstrates a deliberate effort to ensure cross-national perspectives while maintaining a balanced panel composition.

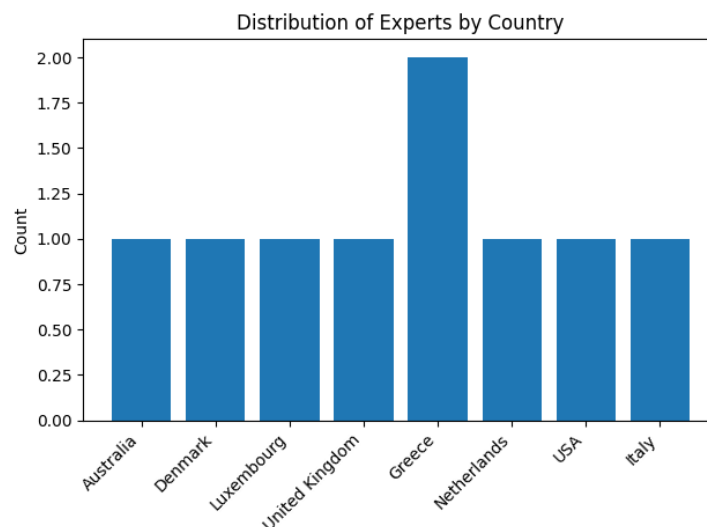


Figure 4 Country-level distribution of experts included in the consultation panel

The predominance of European participants aligns with the regional focus of the project, while the inclusion of experts from the USA and Australia introduces additional external viewpoints,

thereby enhancing the breadth of expertise and supporting the analytical robustness of the consultation.

The distribution of primary areas of expertise among the participating experts, as represented in Figure 5, indicates a strong concentration in disciplines directly relevant to urban resilience and mobility systems. Urban planning and land use, as well as risk management and emergency response, each account for three experts, representing the most prominent knowledge domains within the panel. Transport engineering and mobility systems are represented by two experts, further reinforcing the panel's technical capacity to evaluate infrastructure and operational aspects of resilience. Additionally, one expert reported expertise classified as "other," suggesting complementary knowledge that may contribute to interdisciplinary perspectives. No experts identified environmental science, social sciences, cybersecurity, or policy and governance as their primary domain, indicating that the panel is primarily oriented toward spatial, operational, and risk-related dimensions. Overall, this composition supports a technically grounded evaluation while maintaining a degree of disciplinary diversity appropriate for the study's scope.

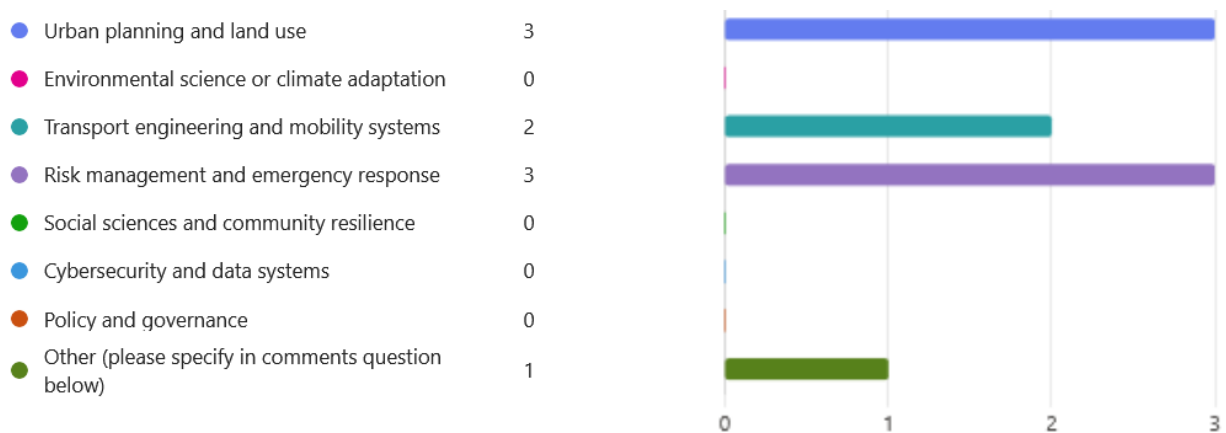


Figure 5 Distribution of primary expertise among panel experts

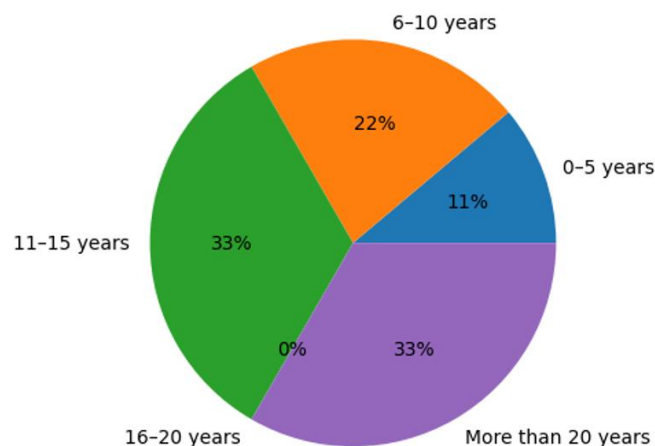


Figure 6 Distribution of experts by years of experience

The distribution of professional experience among the experts is represented in Figure 6, which indicates a panel characterised by substantial seniority and accumulated domain knowledge. Experts with 11-15 years of experience and those with more than 20 years each represent the largest share of the panel (33%), demonstrating a strong presence of highly experienced professionals. Participants with 6-10 years of experience account for 22%, contributing mid-career perspectives that can support balanced and practice-oriented evaluations. Only one expert (11%) reported 0-5 years of experience, while no participants fell within the 16-20-year range.

Overall, the panel composition suggests a high level of expertise, which is particularly desirable in structured expert consultations where informed judgment is critical. The predominance of senior professionals enhances the credibility and analytical robustness of the elicited responses, while the inclusion of less experienced participants introduces complementary viewpoints that may support methodological openness and reduce the risk of overly homogeneous expert opinion. However, as illustrated in Figure 7, the distribution of prior Delphi experience indicates that the majority of participating experts (56%) had not previously taken part in a Delphi consultation, while 33% reported participating once or twice and only 11% had extensive experience with three or more consultations. Although prior Delphi experience can facilitate more structured and confident responses, the inclusion of first-time participants is not considered a limitation in expert elicitation studies, provided that participants possess strong domain expertise. On the contrary, a mix of experienced and novice Delphi contributors may reduce the risk of procedural bias and encourage more independent judgments. The secondary areas of expertise reported by the participating experts further reinforce the interdisciplinary character of the panel. Beyond their primary domains, experts identified complementary knowledge spanning transportation planning, travel behaviour, sustainable mobility, structural engineering, machine learning, climate adaptation, environmental impact assessment, spatial decision-support systems, regional recovery modelling, and environmental design of infrastructure projects. Additional expertise in urban development and regeneration highlights the panel's capacity to address both operational and strategic dimensions of resilient mobility systems.

This breadth of secondary competencies strengthens the analytical foundation of the consultation by integrating perspectives from engineering, data-driven modelling, environmental assessment, and planning. Such interdisciplinarity is particularly valuable in complex decision-making contexts, where the interaction between technological, spatial, and socio-environmental factors must be considered holistically.

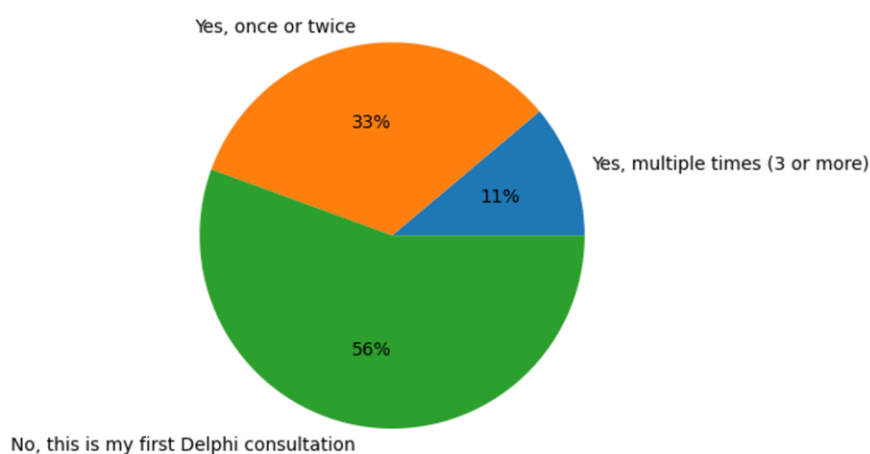


Figure 7 Previous participation in Delphi- consultation

6.3 Analysis of the Delphi-inspired results and prioritisation thresholds

Following data collection, responses from the Delphi-inspired consultation were analysed to validate and prioritise the requirement set using expert agreement as the primary evaluation criterion. Experts rated each requirement using a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree). To measure “agreement” in a transparent and robust way, consensus was assessed through the Interquartile Range (IQR), which quantifies the dispersion of the middle 50% of responses and is widely used in Delphi-type studies due to its resistance to outliers and its intuitive interpretation as concentration of opinion.

A priori acceptance thresholds were defined to translate response dispersion into prioritisation categories. Specifically, requirements were classified as: high consensus when $IQR \leq 1.0$ (indicating that at least half of expert ratings fall within a single scale point), moderate/partial consensus when $IQR = 1.0-2.0$, and low consensus when $IQR > 2.0$, signalling divergence requiring additional clarification and potential reformulation. These thresholds provide a structured bridge between individual expert judgments and traceable decision-support outputs, enabling systematic differentiation between requirements that are already strongly validated versus those needing further deliberation.

The Delphi-inspired round results indicate strong convergence across the panel. Of the 47 analysed requirements, 35 (74.5%) achieved high consensus ($IQR \leq 1.0$), 10 (21.3%) showed moderate consensus (IQR between 1.0 and 2.0), and only one requirement (2.1%) exhibited low consensus ($IQR > 2.0$), as summarised in Table 5.

Table 5 Consensus distribution across requirements

Consensus Level	IQR Threshold	Requirements	Percentage
High consensus	$IQR \leq 1.0$	35	74.5%
Moderate consensus	$IQR 1.0-2.0$	10	21.3
Low consensus	$IQR > 2.0$	1	2.1

In parallel, overall ratings were consistently positive: the mean requirement score was 5.91, the median across requirements was 6.0, and all requirements received mean ratings ≥ 5.0 , indicating broad validation of the co-created requirement set.

This prioritisation was applied as follows: requirements reaching high consensus ($IQR \leq 1.0$; 35 requirements, 74.5%) were interpreted as validated core requirements recommended for retention without modification, forming the stable baseline to inform WP5 SUMA design and implementation. Requirements with moderate consensus ($IQR 1.0-2.0$; 10 requirements, 21.3%), despite positive median ratings, were flagged as conditionally validated, requiring targeted refinement during WP5 implementation activities. The single low-consensus requirement ($IQR > 2.0$; 2.1%) was flagged for re-specification during WP5 refinement activities due to differences in feasibility perceptions, institutional context, or framing ambiguity.

This approach ensures that prioritisation is not based solely on central tendency (e.g., mean or median importance), but on the combination of perceived importance and level of agreement, strengthening the transparency and methodological defensibility of requirement selection. The Delphi-inspired round therefore functions as a structured validation gate, confirming a substantial subset of requirements as stable and widely supported while systematically identifying areas

requiring further clarification. This strengthens the methodological defensibility of the prioritisation process and provides a robust foundation for downstream design and pilot implementation.

6.4 Definition of priority acceptance thresholds for WP5

To ensure that agreement was defined in a transparent and reproducible way, we set acceptance thresholds for each priority before analysing the Delphi-inspired responses. Experts evaluated each requirement using a 7-point Likert scale, and consensus was assessed through the Interquartile Range (IQR), a robust dispersion measure widely applied in Delphi research due to its resistance to extreme values and its capacity to reflect the concentration of expert judgments. Requirements achieving high consensus ($IQR \leq 1.0$) were considered validated core priorities and were therefore accepted as stable inputs to inform WP5 activities without further modification. Requirements demonstrating moderate consensus (IQR between 1.0 and 2.0) were interpreted as conditionally supported; although generally regarded as important, the broader distribution of responses suggested the need for targeted clarification and refinement during WP5 stakeholder iterations before final implementation decisions. Finally, requirements exhibiting low consensus ($IQR > 2.0$) were not immediately accepted as WP5 priorities and were instead flagged for reformulation and additional expert deliberation. Given that overall importance ratings were consistently high across the requirement set, the level of agreement functioned as the principal decision rule for distinguishing validated priorities from those requiring further refinement, thereby strengthening the transparency, traceability, and methodological robustness of downstream decision-making.

Based on the predefined consensus thresholds, 35 statements (as reported in Table 6) achieved high agreement and were treated as core priorities to inform the WP5 baseline requirements specification.

Table 6 Prioritised requirements, with high consensus ($IQR \leq 1.0$) informing WP5.

Requirement No.	IQR	Median	Mean	Title
DEL-R1	1.0	6	6.00	Cybersecurity and Privacy Protection
DEL-R2	1.0	5	5.44	PESTEL Contextual Integration
DEL-R3	1.0	7	6.44	Evacuation and Rerouting Prioritisation
DEL-R4	0.0	6	5.89	Cross-City Knowledge Transfer
DEL-R5	0.0	6	5.89	Urban Planning for Resilient Densification
DEL-R6	1.0	6	5.44	Integration of Public Transport, Pedestrian and Bicycle Data
DEL-R7	1.0	7	6.44	Quick Response to Events
DEL-R8	1.0	6	5.78	Stakeholder Coordination
DEL-R9	1.0	6	6.00	Environment-Friendly Traffic Control
DEL-R10	1.0	6	5.56	Centralised Event Management
DEL-R12	1.0	6	5.89	User-Friendly and Versatile Functions
DEL-R13	1.0	6	6.22	Interdisciplinary Propagation of Disruptive Events
DEL-R14	0.0	6	5.67	Interactions Between Transport Modes
DEL-R15	1.0	6	5.67	AI-Based Predictive Maintenance

DEL-R16	1.0	7	6.33	Multi-Level Coordination and Synchronisation
DEL-R17	0.0	6	5.67	Real-Time Monitoring and Analytics
DEL-R18	1.0	7	6.22	Citizen Mobility Condition
DEL-R19	0.0	6	5.78	Behavioural Nudging for Sustainability
DEL-R20	1.0	6	5.56	Severity and Impact Scales
DEL-R21	1.0	6	6.11	Multi-Modal Data Fusion
DEL-R23	1.0	7	6.11	Resilient Edge Processing Units
DEL-R24	1.0	6	6.22	Economic Impact Modelling
DEL-R25	1.0	6	6.00	Legislation Input
DEL-R28	0.0	7	6.78	Mobility of Emergency Response Services
DEL-R29	1.0	6	6.33	Possible Scenario Evaluation
DEL-R31	1.0	6	6.00	Antifragile Against Disruptions
DEL-R33	1.0	6	6.11	Graph Database Description
DEL-R35	1.0	7	6.67	Geographical Limitations Input
DEL-R37	1.0	6	5.89	Equitable Access and Inclusive Mobility
DEL-R38	0.0	6	5.78	KPI-Driven Performance Monitoring
DEL-R39	1.0	6	5.67	Real-Time Equity Monitoring
DEL-R41	1.0	6	6.44	Simulation of Unexpected Events
DEL-R42	1.0	7	6.56	Flexible Route and Fleet Planning
DEL-R43	1.0	7	6.33	Evaluating Transport Measure Impacts
DEL-R47	1.0	6	6.33	Scenario Analysis and Forecasting System

The single low-consensus requirement identified in the analysis was **DEL-R30 (Usability Across Split Governance)**, which exhibited the highest dispersion of expert ratings (IQR = 3.0), which had an IQR of 3.0, a median of 6, and a mean of 5.56. Expert feedback indicated divergent perspectives primarily related to differences in institutional maturity, governance fragmentation across cities, and uncertainty regarding practical implementation of role-based access control across multi-agency environments. While some experts considered this requirement essential for real-world deployment in complex governance settings, others viewed it as context-dependent or challenging to implement within current pilot capabilities.

Requirements listed in Table 7 achieved moderate consensus (IQR=2.0), indicating moderate dispersion in expert ratings. Divergence across these items primarily reflects differences in perceived implementation maturity, context dependency across cities, and varying interpretations regarding implementation detail versus strategic intent. In particular, several requirements integrate technical functionality with organisational, behavioural, or long-term learning dimensions, which contributed to heterogeneous expert assessments. These requirements are therefore not classified as validated core.

Table 7 Moderate-consensus requirements (IQR = 2.0)

Requirement No.	IQR	Median	Mean	Title
R11	2.0	6	5.44	Real Traffic Data
R22	2.0	6	5.89	Data Provenance and Trust Management
R26	2.0	6	5.89	Antifragility Score Calculation

R27	2.0	6	5.78	Long-Term Transformative Learning
R32	2.0	6	5.78	Prioritisation of Emergencies
R34	2.0	6	5.44	Stakeholder Input
R36	2.0	6	5.56	Learn to Reroute Proactively
R40	2.0	6	5.44	Active Mobility During Disruption
R45	2.0	5	5.67	User Preferences Incorporation During Disruptions
R46	2.0	6	5.11	Energy Optimisation and Emission Reduction Control

7 Requirement Catalogue for SUMA

This section translates the Delphi-informed requirement statements (Table 6) into an implementation-ready requirements catalogue for SUMA. We first provide a background to the process that informed the development of the catalogues. A detailed description of the catalogue is then provided.

Each WP5 baseline requirement (FR/DR/IR/UR/NFR/GR) is derived by consolidating one or more Delphi statements (DEL-R01–DEL-R47). Those Delphi statements are themselves traceable to three prior evidence sources: (i) the systematic literature review and project documentation analysis described; (ii) the ontology competency questions developed in D2.5; and (iii) the structured category refinement undertaken during the General Assembly workshop (Bratislava, December 2025). The ‘Source (Delphi)’ field recorded against each baseline requirement in the catalogue below provides the specific DEL-R link, from which the upstream evidence chain can be followed. Where a baseline requirement consolidates multiple DEL-R statements, this reflects cases where separate Delphi items described the same capability from different stakeholder perspectives.

7.1 Background

The consultation instrument comprised 47 requirement statements (DEL-R01–DEL-R47) presented to experts for rating on a 7-point Likert scale (1 = strongly disagree; 7 = strongly agree). Each statement included an optional free-text field to capture clarification needs, dependencies, and implementation conditions. The full instrument (verbatim statements and response scale) is provided in Appendix 11.

To support implementation and testing, requirements are structured into software-oriented categories. These categories provide a consistent template for specification, prioritisation, and later verification in WP5. Requirements are organised into:

FR Functional requirements

NFR Non-functional requirements

DR Data requirements

IR Interoperability/integration requirements

UR Usability & role-based UI requirements

GR Governance, ethics, privacy requirements

While categories structure requirements for software design and testing, thematic domains provide a complementary coverage check to ensure SUMA supports antifragility not only technically, but also through governance, equity, and long-term learning dimensions.

To complement the technical requirement categories (FR/DR/IR/UR/NFR/GR), requirements are also mapped to thematic urban resilience domains to ensure coverage of the multi-dimensional

nature of antifragile urban mobility. These domains reflect established resilience frameworks and align with AntifragiCity's emphasis on integrating technical performance with governance and social outcomes. The thematic domains are:

1. **Urban Morphology & Land Use Planning**
2. **Environmental Resilience & Climate Adaptation**
3. **Socio-Technical Infrastructure & Mobility Systems**
4. **Risk Governance & Emergency Management**
5. **Social Equity & Community Resilience**

This thematic mapping is used as a coverage check during requirement review and Delphi analysis, ensuring that prioritisation does not over-focus on purely technical requirements at the expense of governance, equity, and long-term learning.

7.2 Requirements catalogue for SUMA

To keep the catalogue directly actionable for WP5 delivery, we distinguish between: (i) WP5 baseline requirements, which are formalised below as uniquely identified FR/DR/IR/UR/NFR/GR requirements with priorities and brief rationales; and (ii) prioritised backlog items, which remain captured as DEL-R statements and will be refined, consolidated, or promoted to the baseline during WP5 iterations where appropriate.

Each WP5 baseline requirement is uniquely identified, categorised (FR/DR/IR/UR/NFR/GR), assigned a priority (Must/Should/Could), and paired with a brief rationale grounded in project objectives and the Delphi-informed validation. Traceability and verification/acceptance criteria for the baseline requirements are provided in Section 11 to ensure each requirement can be tested, inspected, or validated through stakeholder review during WP5. Items not yet promoted to the baseline remain fully documented in Table 3 and are managed through WP5's feedback-driven refinement cycle.

The WP5 baseline requirements below synthesise and consolidate the 47 Delphi statements into an implementable set; multiple DEL-R statements may map to a single WP5 requirement where they describe the same capability from different stakeholder perspectives.

7.3 Functional requirements (FR)

- **FR-01 Disruption/stressor representation (ontology-aligned)**

SUMA shall ingest and represent events/stressors using a consistent taxonomy/ontology so that detection, classification, and scenario generation are comparable across contexts. (WP5 requires an event ontology and data processing layer to map disruptions accurately.)

Rationale: A consistent event/stressor representation is required to compare scenarios across contexts and to support WP5 ingestion, modelling, and analysis services.

Source (Delphi): DEL-R13, DEL-R41

Priority: Must

- **FR-02 Scenario modelling and “what-if” analysis**

SUMA shall support dynamic scenario modelling of urban mobility flows under stress conditions.

Rationale: Scenario and what-if analysis is the core decision-support function of SUMA and underpins WP5 API/UI outputs.

Source (Delphi): DEL-R29, DEL-R41, DEL-R47

Priority: Must

- **FR-03 Resilience and antifragility assessment**

SUMA shall compute and expose resilience/antifragility metrics, including traditional indicators (e.g., congestion, emissions) and measures reflecting the ability to thrive and improve under disruption.

Rationale: Antifragility requires measurable indicators; exposing metrics enables assessment and comparison of strategies and scenarios in WP5.

Source (Delphi): DEL-R26, DEL-R31, DEL-R38

Priority: Must

- **FR-04 Intervention exploration**

SUMA shall allow users to explore alternative interventions and compare expected impacts on resilience/antifragility outcomes under selected scenarios. *(Aligned to WP5 goal of exploring solutions that improve resilience by implementing antifragile attributes.)*

Rationale: SUMA must support exploring interventions to enable decision-makers to compare options and prioritise resilience-building actions.

Source (Delphi): DEL-R24, DEL-R43, DEL-R45, DEL-R46

Priority: Must

- **FR-05 API-first capability for external tooling**

SUMA shall provide a stable, documented API enabling stakeholders to interface proprietary tools and workflows to SUMA services.

Rationale: An API-first architecture enables integration with external tools and supports WP5 interoperability and extensibility goals.

Source (Delphi): DEL-R47, DEL-R30

Priority: Must

- **FR-06 Support near real-time data integration**

SUMA shall support (near) real-time data integration to enable disruption detection and timely scenario updates.

Rationale: Near real-time integration supports timely updates during disruptions and enables operational decision support where data availability permits.

Source (Delphi): DEL-R07, DEL-R17, DEL-R21

Priority: Should

- **FR-07 Learning loop from past disturbances**

SUMA should incorporate mechanisms to learn from historical disruptions/stressors to improve scenario recommendations and performance over time.

Rationale: Capturing lessons and scenario artefacts supports iterative improvement and aligns with antifragility's "learning through stress" principle.

Source (Delphi): DEL-R27, DEL-R04, DEL-R43

Priority: Should

7.4 Data requirements (DR)

- **DR-01 Data source diversity**

SUMA data ingestion should accommodate multiple dynamic streams (e.g., traffic, weather, socio-technical signals) as described in the project's sensing ambitions.

Rationale: Multi-source ingestion is required to represent mobility systems holistically and reflect the project's sensing ambitions.

Source (Delphi): DEL-R06, DEL-R21, DEL-R11

Priority: Should

- **DR-02 Data quality metadata**

SUMA shall store and expose basic metadata (time, provenance, uncertainty/quality flags) to support auditability and decision confidence. *(Supports robust decision support and later evaluation.)*

Rationale: Provenance and quality metadata support trust, auditability, and appropriate interpretation of decision-support outputs.

Source (Delphi): DEL-R22, DEL-R20

Priority: Must

7.5 Interoperability and integration requirements (IR)

- **IR-01 Integration with simulation environments**

SUMA shall be designed to integrate with established tools such as **SUMO**, **PTV Vissim**, and **Aimsun**, with clear interface specifications.

Rationale: Integration with established simulation environments enables realistic scenario execution and accelerates WP5 prototyping and validation.

Source (Delphi): DEL-R41

Priority: Must

- **IR-02 Modular/extensible architecture**

SUMA shall be modular and extensible, allowing new connectors, metrics, and scenario modules to be added without breaking core functionality.

Rationale: Modularity enables incremental delivery and reduces integration risk as new connectors/metrics are added during WP5.

Source (Delphi): DEL-R23, DEL-R33, DEL-R15

Priority: Must

7.6 Usability and role-based UI requirements (UR)

- **UR-01 Role-based access and workflows**

SUMA shall support role-based workflows and interfaces (e.g., planner vs operator vs decision-maker), aligned to WP5's role-based UI deliverable.

Rationale: Role-based workflows reduce complexity and ensure SUMA aligns with stakeholder responsibilities and information needs.

Source (Delphi): DEL-R30, DEL-R16

Priority: Must

- **UR-02 Dashboards and visualisation**

SUMA shall provide interactive dashboards and visualisation layers to communicate scenario results, metrics, and intervention impacts.

Rationale: Visual dashboards are required to communicate scenario results and trade-offs clearly to technical and non-technical users.

Source (Delphi): DEL-R12, DEL-R38, DEL-R39

Priority: Must

7.7 Non-functional requirements (NFR)

- **NFR-01 Documentation quality**

The API shall be well-documented and stable for initial testing, consistent with WP5 outputs.

Rationale: Stable documentation is essential for integration, reproducibility, and effective uptake by WP5 technical and stakeholder users.

Source (Delphi): N/A (engineering requirement)

Priority: Must

- **NFR-02 Performance expectations (near real-time readiness)**

The system should support timely updates consistent with near real-time integration goals.

Rationale: Timely responsiveness is necessary for operational relevance when near real-time feeds are used.

Source (Delphi): DEL-R07, DEL-R29, DEL-R17

Priority: Should

- **NFR-03 Maintainability through agile feedback**

SUMA shall support continuous refinement via agile, feedback-driven development and stakeholder engagement.

Rationale: Feedback-driven maintainability supports iterative refinement and ensures SUMA evolves with stakeholder needs during WP5.

Source (Delphi): DEL-R04, DEL-R27

Priority: Must

7.8 Governance, ethics, privacy (GR)

Because SUMA may ingest socio-technical information and potentially personal data streams, requirements must align with GDPR compliance expectations in project procedures.

GR-01 Data protection and minimisation

SUMA shall enforce data minimisation, role-based access control, and retention rules aligned with GDPR and project data management procedures.

Rationale: SUMA may ingest operational and potentially sensitive data streams; compliance and minimisation reduce deployment risk.

Source (Delphi): DEL-R01 (Cybersecurity and Privacy Protection), DEL-R30 (Usability across split governance).

Priority: Must

GR-02 Auditability and accountability

SUMA shall maintain audit trails for critical actions (scenario runs, parameter changes, data injections, decision outputs), including who performed the action and when.

Rationale: Traceable decision-support is necessary for institutional accountability and post-event review.

Source (Delphi): DEL-R22 (Data provenance and trust management), DEL-R34 (Stakeholder input with audit trails).

Priority: Must

GR-03 Transparency of assumptions and limitations

SUMA shall expose key assumptions, data quality/provenance indicators, and uncertainty/limitations alongside outputs to mitigate misinterpretation and misuse.

Rationale: Antifragility assessment requires honest communication of uncertainty and context-dependence.

Source (Delphi): DEL-R20 (Severity and impact scales), DEL-R22 (Provenance/trust).

Priority: Must

GR-04 Secure stakeholder input governance

SUMA shall provide controlled mechanisms for authorised stakeholders to inject local constraints and emergent risks, with validation rules and governance workflows.

Rationale: Local knowledge improves operational relevance but requires safeguards.

Source (Delphi): DEL-R34 (Stakeholder input).

Priority: Should

8 Requirements management and traceability

To ensure that each requirement directly informs WP5 implementation, we define a traceability structure linking requirements to WP5 deliverables (D5.1–D5.4), verification methods, and acceptance criteria that can be evaluated during development and stakeholder review.

To support WP5 implementation and testing, requirements in this deliverable are not only mapped to WP5 outputs but are also expressed in a form that can be verified. Where available, the competency-question (CQ) approach used in the upstream ontology requirements work is treated as a practical bridge from “requirement intent” to “testable specification”: CQs translate high-level requirements into concrete queries the system must answer, which can then be implemented as inspection points, API tests, or scenario-based checks during WP5 integration.

8.1 Traceability matrix

Each **WP5 baseline requirement** is traced to:

- the relevant **WP5 outputs/deliverables** (D5.1-D5.4) and the associated SUMA component(s) (e.g., ontology/data layer, simulation integration, metrics, API architecture, UI/workflows),
- an explicit **verification method** (test, demo, inspection, stakeholder review), and
- a concise **acceptance criterion** describing the evidence needed for WP5 acceptance.

Scope of traceability in this deliverable. Table 8 reports traceability for the **WP5 baseline requirements** formalised (FR/DR/IR/UR/NFR/GR). The full set of elicited Delphi statements (DEL-R01...DEL-R47) is documented in Table 3 and remains traceable via the “Source (Delphi)” links; statements not yet promoted to the WP5 baseline are managed as prioritised backlog items and may be consolidated or added to the baseline during WP5 iterations. The traceability matrix is presented in Table 8.

Table 8 Traceability matrix

Req ID	Category	Source (Delphi)	WP5 output(s)	Verification method	Acceptance criteria (summary)
FR-01	FR	DEL-R13, DEL-R41	D5.1, D5.2	Test inspection +	Method: Test + inspection. Given an ontology-aligned schema for event/stressor objects, when a user/API submits a valid event instance, then the instance is accepted, stored, and retrievable, and schema validation passes; when an invalid instance is submitted, then it is rejected with a validation error. Evidence: schema validator output + API responses + stored event record + one end-to-end scenario run log referencing the event ID.
FR-02	FR	DEL-R29, DEL-R41, DEL-R47	D5.1, D5.2, D5.3	Test + demo	Method: Test + Demo. The test component checks that SUMA returns comparable outputs for two scenario configurations (baseline and disrupted) via both the UI and the API, using the same KPI schema and units, with scenario identifiers and timestamps present in the response. This component produces automated evidence: API response payloads, scenario execution logs, and UI export files. The demo component is a structured walkthrough in which a representative user navigates the comparison view and confirms it is intelligible. Reviewer observations are recorded in session notes, which form part of the acceptance record alongside the test artefacts.
FR-03	FR	DEL-R26, DEL-R31, DEL-R38	D5.1, D5.3	Test stakeholder review +	Method: Test + stakeholder review. Given a scenario run with defined KPI/metric configuration, when SUMA computes resilience/antifragility metrics, then metrics are displayed and exportable with their definitions (name, formula/reference, input variables, units) available in documentation or UI help. Evidence: metrics export (CSV/JSON) + UI screenshots + documentation excerpt; stakeholder

					review minutes confirming interpretability for defined roles.
FR-04	FR	DEL-R24, DEL-R43, DEL-R45, DEL-R46	D5.2, D5.3	Test + stakeholder review	Method: Test + stakeholder review. Given at least two intervention options (e.g., policy/control/routing) applied to the same baseline, when the user executes scenarios, then outputs are generated for each intervention and a comparison view/export shows trade-offs across selected KPIs (including environment/equity where configured). Evidence: scenario configurations + output exports + comparison view screenshots; stakeholder review notes confirming usefulness.
FR-05	FR	DEL-R47, DEL-R30	D5.1, D5.2	API test	Method: API test. An API test sends HTTP requests to documented SUMA endpoints using defined input payloads and verifies that: (i) valid requests from authenticated clients are accepted, processed, and return results through versioned, documented endpoints; (ii) responses conform to the published OpenAPI schema; and (iii) unauthenticated or unauthorised requests are rejected with the appropriate status codes. Evidence consists of the automated test report, schema validation output, and example request/response logs. This is distinct from a UI-level demo or a manual inspection.
FR-06	FR	DEL-R07, DEL-R17, DEL-R21	D5.2, D5.3	Performance test + demo	Method: Performance test + demo. Given a configured near-real-time ingestion mode and a reference input stream, when new data/events are ingested, then SUMA updates the relevant indicators and dashboards/API outputs within the target latency defined for WP5 testing, and the update is traceable in system logs. Evidence: ingestion timestamps + processing logs + UI update capture or API polling trace demonstrating end-to-end update timing.

FR-07	FR	DEL-R27, DEL-R04, DEL-R43	D5.3, D5.4	Inspection + stakeholder review	Method: Inspection + stakeholder review. Given completed scenario runs, when a user captures a lesson/annotation or stores a scenario artefact, then the artefact is persisted, searchable, and retrievable with metadata (scenario ID, author/role, timestamp), and can be referenced in subsequent iterations. Evidence: artefact record screenshots/export + search/retrieval demonstration + stakeholder review notes confirming it supports iterative refinement.
DR-01	DR	DEL-R06, DEL-R21, DEL-R11	D5.1, D5.2	Inspection + test	Method: Inspection + test. Given documented data schemas/connectors, when representative datasets for at least two modes (e.g., road traffic + public transport, plus active modes where available) are ingested, then SUMA stores them using the defined data model and exposes them to scenario configuration without schema errors. Evidence: data schema documentation + ingestion logs + API/UI evidence of datasets available for scenario setup.
DR-02	DR	DEL-R22, DEL-R20	D5.1, D5.2	Inspection + test	Method: Inspection + test. Given an ingested dataset, when SUMA stores and uses it, then provenance fields (source, timestamp, version/licence where applicable) and quality/uncertainty indicators are recorded and returned with outputs; if input quality is missing/low, the UI/API flags it visibly. Evidence: stored metadata record + API output payload showing provenance/quality fields + UI screenshot of flags + test case demonstrating missing/low-quality flagging.
IR-01	IR	DEL-R41	D5.2	Demo	Method: Demo. A configured end-to-end integration run using at least one external simulator connector (e.g., SUMO) is executed and observed. SUMA sends a scenario configuration to the external simulator, receives simulation outputs, and stores them under the

					correct scenario ID for downstream visualisation and analysis. The run is logged throughout. Evidence required for acceptance: the connector configuration file, the simulator run log, the returned output files or records, and the SUMA scenario record confirming the imported results are accessible. Reviewer observations during the session are recorded in session notes.
IR-02	IR	DEL-R23, DEL-R33, DEL-R15	D5.2, D5.4	Inspection + regression test	Method: Inspection + regression test. Given a defined module/interface contract and versioning policy, when a new module/connector is added or updated, then existing APIs remain backward-compatible (or versioned appropriately) and the existing test suite passes. Evidence: interface documentation + versioning notes + regression test report demonstrating no breakage of existing endpoints/workflows.
UR-01	UR	DEL-R30, DEL-R16	D5.3	Stakeholder review + test	Method: Stakeholder review + test. Given at least two defined roles (e.g., planner and operator), when each role logs in and performs their workflow, then the system enforces role-based permissions (allowed/denied actions) and provides the role-relevant navigation/views. Evidence: RBAC test cases + access-denied logs + role-specific UI screenshots; stakeholder review minutes confirming workflow fit.
UR-02	UR	DEL-R12, DEL-R38, DEL-R39	D5.3	Stakeholder review	Method: Stakeholder review. Given at least two scenario runs, when a user opens the dashboard, then the dashboard presents side-by-side comparison for selected KPIs and (where configured) distributional/equity indicators with clear labels, units, and scenario identifiers. Evidence: dashboard screenshots/export + stakeholder review feedback confirming clarity and usefulness.

NFR-01	NFR	N/A (engineering requirement)	D5.1, D5.3	Inspection	Method: Inspection. Given the delivered WP5 artefacts, when a new technical user follows the documentation, then they can install/access SUMA, call at least one API endpoint, execute a minimal scenario workflow, and interpret the returned outputs using the documented schemas and definitions. Evidence: versioned documentation (API + UI) + a reproducibility checklist or walkthrough log.
NFR-02	NFR	DEL-R07, DEL-R29, DEL-R17	D5.2	Performance test	Method: Performance test. Given a reference workload (scenario/stream) and an agreed performance target for WP5 testing, when the system is exercised under that workload, then throughput/latency measurements are produced and meet the target; if not, the system's degraded-mode behaviour is documented and demonstrable. Evidence: performance test report + monitoring logs + degraded-mode specification and demonstration notes.
NFR-03	NFR	DEL-R04, DEL-R27	D5.4	Inspection + stakeholder review	Method: Inspection + stakeholder review. Given the WP5 development process, when changes are proposed from stakeholder feedback, then they are logged, prioritised, and tracked through an iteration cycle with status and resolution recorded. Evidence: issue tracker extracts or change log + iteration notes + stakeholder validation confirming feedback incorporation.
GR-01	GR	DEL-R01, DEL-R30	D5.2, D5.4	Inspection	Method: Inspection. Given governance/privacy requirements, when roles and retention policies are configured, then access control, minimisation, and retention settings are enforceable and documented, and privacy notes (e.g., processing purposes, data types, retention) are included in system documentation. Evidence: configuration screenshots + policy documentation + access control specification.

GR-02	GR	DEL-R22, DEL-R34	D5.2, D5.4	Inspection + test	Method: Inspection + Test. The inspection component consists of a structured review of audit log records produced during a defined set of critical actions (scenario run, parameter change, data injection, export). A reviewer examines the records and confirms that each entry contains the required fields: user/role, timestamp, action type, scenario ID, and outcome. The reviewer also confirms that logs can be retrieved and exported for external review. Evidence: annotated log extracts and a written review record. The test component executes each critical action category programmatically and checks that a corresponding audit log entry is created with the required fields. Evidence: automated test report and log extract showing coverage of each action type.
GR-03	GR	DEL-R20, DEL-R22	D5.3, D5.4	Stakeholder review + inspection	Method: Stakeholder review + inspection. Given a scenario output view/export, when results are presented, then the output includes explicit assumptions/limitations and provenance/quality indicators, and uncertainty cues where applicable; reviewers confirm these elements are sufficient to reduce misinterpretation. Evidence: output export/UI screenshots showing fields + stakeholder review minutes.
GR-04	GR	DEL-R34	D5.2, D5.4	Test inspection +	Method: Test + inspection. Given an authorised stakeholder input workflow, when a stakeholder submits a new constraint/risk input, then the system validates it against defined rules, records provenance (who/when/what), routes it through the configured governance workflow (approve/reject), and includes it (if approved) in subsequent scenario configurations with traceability. Evidence: submission record + validation outcome + approval log + scenario configuration showing applied input.

8.2 Acceptance criteria examples

The following sub-section provides examples of the acceptance criteria that may be used when implementing the proposed requirements:

- **FR-05 pass:** external tool can call API endpoints to submit scenario parameters and retrieve metrics with documented schemas.
- **IR-01 pass:** at least one integration pathway demonstrated (e.g., SUMO connector prototype) consistent with integration intent.
- **UR-02 pass:** dashboards display scenario comparisons and antifragility/resilience metrics clearly for a defined role.
- **GR-01 pass:** access control rules are enforced for each role; retention policies are configurable; privacy impact notes are included in system documentation.
- **DR-02 pass:** outputs include provenance references to datasets and timestamps; missing/low-quality inputs are flagged in UI/API responses.
- **NFR-02 pass:** for a reference scenario, end-to-end update latency remains within the target window defined by the selected deployment mode (batch/near-real-time).

8.3 Assumptions, constraints, and open issues

The following sub-section summarises (a) the acknowledged assumptions in relation to the captured requirements, (b) the potential constraints that should be considered, and (c) the open issues to be resolved when undertaking the implementation.

Assumption:

- A1. SUMA will be delivered as a modular environment with an API-first core (D5.1) and optional connectors/integrations (D5.2).
- A2. SUMA requirements prioritisation reflects expert consensus on importance, not a claim of feasibility in every city context; feasibility is handled through WP5 iterative refinement.
- A3. No pilot performance results are reported here beyond what is required to justify requirements; validation in this deliverable focuses on expert agreement and traceability.

Constraints:

- C1. Data availability and licensing constraints may limit access to real-time feeds; SUMA must support degraded modes and synthetic/test data.
- C2. Integration constraints vary across simulation tools (SUMO/Vissim/Aimsun); connectors may be staged incrementally in WP5.
- C3. Governance and privacy constraints differ by city and operator; role-based access and auditability are mandatory design constraints.

Open issues:

- O1. Definition of target latency windows per deployment context (control room vs planning) and the associated infrastructure requirements.
- O2. Formal selection of baseline KPI set and antifragility score operationalisation for each pilot context.
- O3. Agreement on minimum interoperability scope for D5.2 (which tools/connectors are “must-have” vs “nice-to-have”).
- O4. Finalisation of user role model and permissions matrix for D5.3 dashboards and workflows.

9 Discussion

This deliverable addresses five research questions to ensure SUMA requirements are both implementable within WP5 and traceable to verification activities without reporting pilot results. The results show that the adopted requirements elicitation process provided a structured way to translate antifragility concepts into concrete system requirements for urban mobility applications. The refinement of analytical categories and their subsequent evaluation by experts indicate consistency between insights from the literature and the practical needs identified by project partners and pilot cities. The prioritised requirements emphasise capabilities related to adaptive decision support, scenario analysis, data management, interoperability, and user interaction, suggesting that antifragility is supported by the combined functioning of multiple system components rather than by a single technical feature. The use of a modified Delphi approach proved suitable for validating requirements in a context where empirical evidence is limited and implementation constraints are present, allowing informed expert judgment to guide prioritisation in a transparent manner. Differences in expert agreement across certain requirements reflect the diversity of operational contexts and levels of maturity rather than deficiencies in the methodology, indicating areas where further clarification or iteration may be needed during later project stages. Although the expert panel size was limited, the prior integration of literature review, project documentation, and stakeholder input strengthened the relevance and coherence of the requirement set. Overall, the results provide a stable and structured basis for subsequent system design activities, while allowing flexibility for refinement as implementation and pilot testing progress.

The prioritised requirements will serve as foundational inputs for subsequent WP5 technical development and integration activities. These prioritised Delphi statement IDs (R/DEL-R) are subsequently consolidated and formalised into WP5 baseline requirement IDs (FR/DR/IR/UR/NFR/GR) in Section 8 to ensure implementability, avoid duplication, and enable verification/acceptance planning.

Each requirement in the catalogue is specified with an acceptance criterion designed to be verifiable during WP5 development and integration. Verification methods are selected to match requirement type: (i) functional requirements are verified via scenario-based test cases and expected simulator outputs; (ii) data and interoperability requirements are verified through interface conformance checks (schema/API validation), data quality checks, and successful end-to-end data exchanges between components; (iii) usability requirements are verified through structured user tasks with measurable completion and satisfaction criteria; and (iv) non-functional and governance/privacy requirements are verified through security/privacy checks (e.g., access control tests, logging/auditability checks, and evidence of compliance documentation). This ensures that the elicited requirements remain not only prioritised, but also testable and implementable within WP5.

The resulting prioritised statements are then formalised into a WP5-oriented requirements catalogue, structured by category (functional, data, integration, usability, non-functional, governance) and paired with verification and acceptance criteria.

RQ1: Who are the intended users and stakeholders of SUMA, and what decisions and workflows must the system support across planning, operations, and emergency contexts?

RQ2: Which functional capabilities must SUMA provide to represent stressors, run scenario-based analyses, and assess resilience/antifragility in a way that is implementable within WP5?

RQ3: What data, interoperability, and integration requirements are necessary for SUMA to operate with heterogeneous sources and connect to external simulation environments and tools?

RQ4: Which non-functional, usability/UI, and governance/privacy requirements are essential to ensure SUMA is usable, trustworthy, secure, and compliant in real-world institutional settings?

- **RQ5:** How are requirements structured, linked to WP5 deliverables (D5.1–D5.4), and expressed with verification criteria to support implementation planning?

RQ1 (Who are the intended users and stakeholders of SUMA, and what decisions and workflows must the system support across planning, operations, and emergency contexts?): SUMA is framed as an urban mobility decision-support environment for heterogeneous stakeholders, including city decision-makers, transport planners and operators, emergency management actors, platform/API integrators, and researchers. These groups imply distinct workflows, strategic planning, operational monitoring, crisis coordination, and tool integration, requiring role-based access, tailored dashboards, and auditable decision outputs.

RQ2 (Which functional capabilities must SUMA provide to represent stressors, run scenario-based analyses, and assess resilience/antifragility in a way that is implementable within WP5?): The requirement set converges on core functional capabilities, representing disruptions/stressors consistently, performing scenario-based “what-if” analysis, computing resilience and antifragility metrics, exploring interventions with trade-off visibility, and supporting learning loops from past disturbances. These functions reflect an antifragile orientation by extending beyond robustness and recovery to include mechanisms through which systems can learn from disruptions and enhance performance over time.

RQ3 (What data, interoperability, and integration requirements are necessary for SUMA to operate with heterogeneous sources and connect to external simulation environments and tools?): Requirements emphasise multi-modal data coverage, provenance/quality metadata, and the ability to connect to external simulation environments and tools through stable interfaces. Interoperability requirements are particularly central because SUMA is intended to be compatible with established transport simulators and also usable in standalone modes. The catalogue therefore treats connectors and data governance as first-class design concerns rather than implementation afterthoughts.

RQ4 (Which non-functional, usability/UI, and governance/privacy requirements are essential to ensure SUMA is usable, trustworthy, secure, and compliant in real-world institutional settings?): Expert validation strongly supports cybersecurity, privacy safeguards, auditability, and transparency as deployment preconditions. Non-functional requirements also reflect maintainability and iterative refinement through agile feedback loops, consistent with WP5’s planned stakeholder engagement. Usability requirements stress role-based workflows and accessible communication during crisis contexts, acknowledging that decision-support quality depends on interpretability and trust as much as technical correctness.

RQ5 (How are requirements structured, linked to WP5 deliverables (D5.1–D5.4), and expressed with verification criteria to support implementation planning?): Requirements are organised into six categories (FR/DR/IR/UR/NFR/GR) and prioritised using an IQR-based consensus rule from the Delphi-inspired consultation, which identifies a stable core set and flags context-sensitive items for refinement. Each requirement is linked to specific WP5 deliverables (D5.1–D5.4) and paired with a

verification method and acceptance criterion in the traceability matrix (Table 8), enabling requirements to directly drive implementation backlogs and testing plans.

Overall, the requirements catalogue provides a structured bridge from project objectives and expert validation to WP5 implementation planning, while keeping the deliverable firmly within the scope of requirements elicitation and specification.

10 Conclusion

This deliverable formalises a validated and prioritised requirements baseline for SUMA (WP5/KER5) and provides the structure needed to implement and verify the system in D5.1–D5.4. Addressing RQ1, we define SUMA’s intended users and stakeholders and the decision-support workflows that motivate role-based access, dashboards, and auditable outputs. Addressing RQ2, we specify the core functional capabilities required to represent stressors, run scenario analyses, compute resilience/antifragility metrics, explore interventions, and support learning loops consistent with antifragile system behaviour. Addressing RQ3, we define data and interoperability requirements that enable multi-modal data fusion, provenance and trust management, and staged integration with established simulation tools. Addressing RQ4, we codify non-functional, usability, and governance/privacy requirements, including cybersecurity, GDPR-aligned controls, and transparency, recognising these as necessary conditions for trustworthy deployment. Finally, addressing RQ5, we provide a prioritisation through an expert consultation process and deliver a traceability and acceptance framework that links each requirement to WP5 outputs and verifiable criteria.

The resulting specification will directly feed into the implementation and testing of the SUMA platform, while leaving contextual feasibility decisions and refinements to the planned agile feedback loops and stakeholder engagement activities.

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12 Appendix

A) Delphi consultation and in-depth literature review.

To situate the methodological choices of this study, this section reviews the Delphi method and its application in requirements elicitation for complex socio-technical systems. Particular attention is given to methodological adaptations, including modified and context-specific Delphi approaches used to validate analytically derived requirement sets. This review provides the foundation for the Delphi-inspired expert consultation implemented in AntifragiCity.

12.1.1 Delphi Method Fundamentals

The Delphi technique originated at the RAND Corporation in the 1950s as a systematic forecasting method using expert judgment to address complex problems lacking definitive solutions [37], [42]. The method derives its name from the Oracle of Delphi, symbolizing its purpose of obtaining reliable consensus through structured expert consultation [38], [43]. At its core, Delphi is characterized by three fundamental principles: anonymity of participants to reduce social pressure and groupthink; iterative rounds with controlled feedback allowing experts to reconsider positions; and statistical aggregation of group responses to achieve consensus [39], [42].

The method has evolved into several variants adapted to different research contexts. Classical Delphi maintains strict anonymity and typically runs three to four rounds until stability is achieved [36], [62]. Modified Delphi begins with pre-structured questions rather than open-ended exploration, reducing time and resources while maintaining rigor [37], [40]. Policy Delphi explicitly seeks to explore divergent perspectives rather than consensus, useful for contested policy domains [10], [63]. Real-time Delphi leverages digital platforms to provide immediate feedback, collapsing traditional rounds into continuous consultation [64], [65].

Delphi is preferred over alternative consensus methods when the problem requires subjective judgment from dispersed experts, lacks sufficient historical data for quantitative forecasting, or benefits from anonymity to prevent dominance by high-status individuals [36], [43]. Compared to focus groups, Delphi eliminates geographic constraints and social dynamics that bias face-to-face deliberation [38], [63]. Unlike the Nominal Group Technique (NGT), which converges in a single session, Delphi's iterative structure allows reflection and reconsideration, particularly valuable for complex socio-technical systems [36], [37]. Cerè et al. [53] illustrate how a modified Delphi can validate and prioritise a pre-structured set of criteria in an applied engineering context under practical constraints, combining transparent consensus reporting with iterative refinement across rounds. In AntifragiCity, we draw on this general logic of structured expert consultation and transparent agreement reporting but adapt it to a Delphi-inspired single-round prioritisation step whose purpose is to produce an implementable WP5 requirements baseline rather than to run a full classical Delphi termination process.

12.1.2 Requirements Identification through Delphi

Delphi has proven effective for requirements elicitation across diverse domains, particularly where stakeholder knowledge is tacit, distributed, or contested [45]. In software engineering, Delphi supports requirements prioritization when multiple stakeholders hold conflicting needs [36]. In healthcare, Delphi elicits quality indicators and clinical guidelines where evidence is incomplete [38],

[48]. The method's structured anonymity prevents premature closure around dominant voices, enabling comprehensive requirements capture [44], [45].

Question formulation is critical to Delphi success. Best practices emphasize clarity, specificity, and avoidance of double-barreled questions that conflate multiple issues [38], [63]. Questions should be unambiguous, avoiding jargon unless universally understood by the panel [38]. Rating scales (Likert 1-5 or 1-7) are preferred over binary yes/no formats to capture nuance and enable statistical analysis [39], [41]. The guidance paper [53] structured questions around **seven thematic clusters** (environmental, socio-organizational, technical) with Likert-scale importance ratings, allowing both within-category consensus measurement and cross-category comparison. This clustered approach aligns with recommendations for complex systems assessment where interdependencies across categories require holistic evaluation [36], [37].

Expert panel selection balances disciplinary diversity against shared understanding of the problem domain [66]. Panels typically range from 10 to 50 participants, with diminishing returns beyond 20-30 for homogeneous groups [43]. Heterogeneous panels addressing multi-stakeholder systems may require 30–50 to ensure representativeness [67]. Selection criteria include: domain expertise (publications, professional experience); stakeholder representation (balanced across affected constituencies); and commitment to participate through multiple rounds [42]. Cerè et al. [53] recruited **23 first-round and 21 second-round participants** from academia, industry, and government, ensuring multi-disciplinary coverage of building resilience assessment, with a panel size consistent with best-practice recommendations for applied engineering studies [37].

Iterative rounds and convergence criteria define Delphi's dynamic structure. Most studies employ 2-4 rounds, with three being most common [43]. Round 1 may be exploratory (open-ended) or structured (pre-defined criteria for validation); subsequent rounds present aggregated feedback and request re-rating [45]. Convergence is assessed through statistical measures: coefficient of variation ($CV < 0.5$), interquartile range ($IQR \leq 1$ on 5-point scales), or percentage agreement (≥ 70 -75%) [41]. The guidance paper employed **two rounds**, using percentage agreement and descriptive statistics to validate criteria, a pragmatic choice given the applied context and expert time constraints [53]. This two-round design is increasingly common in policy-oriented Delphi studies where rapid turnaround is required [48].

12.1.3 Methodological Considerations

Questionnaire design must balance comprehensiveness with respondent burden. Instruments typically begin with demographic profiling, followed by core rating tasks, and conclude with optional open-ended feedback [38]. Instructions should clarify the rating scale, define technical terms, and explain how feedback will be used in subsequent rounds [39]. Pilot testing with 3-5 non-panel experts identifies ambiguities before full deployment [36]. Cerè et al. [53] structured their instrument around **seven criterion categories** with standardized Likert scales, using MATLAB for statistical analysis. This demonstrates rigorous quantitative treatment of qualitative expert judgment.

Anonymity versus interaction represents a key design choice. Traditional Delphi maintains complete anonymity to prevent authority bias and social conformity [36], [37]. However, modified approaches allow limited interaction, such as revealing individual positions without identity [43], [45]. Real-time Delphi platforms provide immediate aggregated feedback, creating partial transparency [64]. The guidance paper maintained **controlled anonymity**, presenting aggregated Round 1 results to Round 2 participants without disclosing individual responses, balancing transparency with independence [53].

Consensus measurement employs multiple statistical indicators. Descriptive statistics (mean, median, mode) summarize central tendency, while dispersion measures (standard deviation, IQR, CV) quantify agreement [41]. Consensus thresholds vary by domain: medical guideline studies often require $\geq 80\%$ agreement, while policy studies accept $\geq 60\text{--}70\%$ [48]. In multi-round Delphi studies, researchers [41], [45] may also assess stability by examining whether responses change materially between rounds; some authors caution that consensus (e.g., IQR) alone is not always sufficient as a stopping criterion because ratings may still fluctuate across rounds [53]. For example, hierarchical stopping criteria discuss stability checks alongside consensus, including comparing variation-based indicators between rounds. However, such round-to-round stability criteria are not applicable to a single-round design. Accordingly, in D2.6 we use within-round dispersion (e.g., IQR) to report agreement transparently for prioritisation, and we treat lower-agreement items as candidates for refinement during WP5's iterative stakeholder cycles rather than claiming Delphi "stability".

In applied, socio-technical projects, the primary methodological goal is often to capture and formalise implementable requirements rather than to pursue "classical Delphi" as an end in itself. For this reason, Delphi-like and other structured expert elicitation variants are commonly adapted to project constraints, provided that independence of judgement, transparency of the aggregation rule, and clear documentation of scope and outputs are maintained. Related work on robust requirements gathering in complex urban cyber-physical domains emphasises that requirement quality depends on balancing multiple perspectives (domain relevance, technical feasibility/integration, and stakeholder ownership), and that "robustness" is strengthened when structured expert input is combined with complementary sources (e.g., documentation review and workshops) and when requirements are made testable through explicit validation/verification hooks. These principles motivate our Delphi-inspired single-round design as a pragmatic and defensible approach for prioritising a pre-structured catalogue under time constraints while preserving transparency and traceability to downstream implementation.

Validity and reliability are enhanced through transparent process documentation and triangulation with other evidence sources [45]. Content validity is strengthened by systematic panel selection across relevant expertise domains [48]. Construct validity requires alignment between Delphi criteria and established theoretical frameworks [66]. Test-retest reliability can be assessed by repeating final-round items with a subset of panelists [38]. The guidance paper grounded criteria in existing resilience literature and disaster management frameworks, enhancing content validity; the two-round structure with statistical convergence analysis provided reliability [53]

12.1.4 Practical Implementation

Online versus traditional Delphi involves trade-offs between efficiency and depth. Digital platforms (SurveyMonkey, Qualtrics, dedicated Delphi software) reduce administration burden, enable real-time tracking, and facilitate data analysis [65]. However, they may limit rich qualitative feedback compared to paper-based or interview approaches [38]. Hybrid designs combine structured online ratings with follow-up interviews to explore divergent views [39]. Cerè et al. [53] do not specify the delivery mode but demonstrate **MATLAB-based statistical analysis**, suggesting digital data collection to enable computational processing.

Timeline and resource requirements depend on round number, panel size, and response rates. A three-round Delphi typically requires 3-6 months: 3-4 weeks per round for response collection, 1-2 weeks for analysis and feedback preparation [37]. Attrition between rounds averages

20-30%; studies should over-recruit by 250-40% to maintain statistical power [62]. Resource needs include survey platform costs, statistical software, and researcher time for qualitative coding and feedback synthesis [36]. The guidance paper's **two-round design** with 23 and 21 participants demonstrates pragmatic resource management, accepting modest attrition (8.7%) while achieving robust validation across 43 final criteria [53].

Common challenges include low response rates, panel fatigue across rounds, and managing divergent opinions that resist convergence [43]. **Mitigation strategies** include: personalized recruitment emphasizing study importance and expert contribution; limiting questionnaire length (≤ 30 minutes per round); providing timely, meaningful feedback showing how inputs shaped subsequent rounds; and using adaptive designs that reduce rounds for stable items [41]. When consensus remains elusive, researchers should document persistent disagreement rather than forcing artificial convergence, as divergence may signal legitimate conceptual debate [41].

Ethical considerations include informed consent, confidentiality protection, and managing expectations about study influence on policy [67]. Participants should understand time commitments, how anonymity will be preserved, and whether findings will be disseminated [38]. Transparency about limitations, such as inability to guarantee policy adoption maintains trust and reduces participant disengagement [45].

12.2 Application Domains: Urban and Mobility Systems

Delphi has been extensively applied to **urban planning and transportation**, where complex socio-technical systems, multiple stakeholder perspectives, and uncertain futures favour expert judgment methods. In **urban resilience assessment**, Delphi has elicited indicators for community disaster preparedness [52], validated smart city performance metrics [51] and prioritized green infrastructure investments [50]. The guidance paper [53] extends this tradition to **building-scale resilience**, demonstrating how Delphi can operationalize abstract resilience concepts into measurable criteria across environmental, technical, and socio-organizational dimensions, as a transferable logic for urban mobility systems.

In **transportation and mobility**, Delphi studies have forecasted autonomous vehicle adoption timelines [68], identified barriers to sustainable mobility transitions (Hickman and Banister, 2014), and validated freight resilience indicators [54]. These applications share methodological adaptations suited to urban systems: (1) **multi-stakeholder panels** balancing technical experts, policymakers, and community representatives; (2) **iterative refinement** of indicators to capture cross-scale dynamics (individual travel behaviour $\rightarrow$ network performance $\rightarrow$ regional equity); and (3) **integration with quantitative models**, using Delphi to parameterize simulation models or validate data-driven analytics [44].

Methodological adaptations across domains reveal domain-specific best practices. Medical Delphi studies emphasize stringent consensus thresholds ($\geq 80\%$) given clinical stakes, whereas urban planning studies accept lower thresholds ($\geq 60\text{--}70\%$) reflecting legitimate policy pluralism [48]. Engineering applications favour **modified Delphi with pre-structured criteria** to enable quantitative analysis, as demonstrated by Cerè et al. [53]. Policy-oriented studies in urban governance often employ **policy Delphi** to map contested viewpoints rather than force consensus [10]. The guidance paper's **seven-category clustering** (environmental, socio-organizational, technical) mirrors best practices in complex systems assessment, ensuring comprehensive coverage while enabling statistical comparison across dimensions [36].

Critical success factors identified across domains include: (1) **Early stakeholder engagement** to build legitimacy and ensure panel commitment [45] (2) **Theoretical grounding** linking Delphi

criteria to established frameworks, as Cerè et al.[53] did with disaster management and resilience literature; (3) **Mixed-methods integration**, combining Delphi with quantitative modelling, case studies, or scenario analysis [37] and (4) **Actionable outputs**, translating consensus into decision frameworks, as the guidance paper did by producing a **43-criteria assessment framework** ready for urban-scale application [53].

Transferability to urban mobility research: The guidance paper's approach offers several transferable elements for AntifragiCity and similar projects. First, its **two-round modified Delphi** balances rigor with pragmatism, suitable for time-constrained multi-partner projects. Second, **thematic clustering** (seven categories in Cerè et al [53]) can structure urban mobility requirements around the five-pathway taxonomy (Physical, Informational, Behavioural, Economic, Institutional). Third, **statistical validation** using percentage agreement and mean importance scores provides transparent criteria for inclusion/exclusion, adaptable to competency question validation. Fourth, **iterative expansion** (40 → 48 → 43 criteria) demonstrates the method's capacity to incorporate emergent expert insights while maintaining rigor, which is valuable for exploratory requirements gathering in novel domains like antifragile mobility systems.

12.3 Delphi instruments and reporting

12.3.1 Requirement statement questionnaire template (Round 1 – single round)

The questionnaire comprised the complete requirement set (47 statements; in Table 3). Each requirement was presented as a single statement written in “shall” form, followed by a structured rating item and an optional comment box.

Rating item (mandatory): “To what extent should this requirement be retained as a core SUMA requirement for WP5 development?”

Scale: 1 = Strongly disagree; 2 = Disagree; 3 = Somewhat disagree; 4 = Neither agree nor disagree; 5 = Somewhat agree; 6 = Agree; 7 = Strongly agree.

Comment box (optional): “If you selected 1–4, or if you believe the statement is ambiguous/context-dependent, please explain your reasoning and propose improved wording.”

12.3.2 Post-consultation feedback summary (single-round)

After Round 1 data collection, a short feedback summary was prepared for internal reporting and WP5 refinement. The summary includes: (i) descriptive statistics per requirement (median, IQR, mean), (ii) consensus classification (high/moderate/low), and (iii) a thematic synthesis of expert comments highlighting ambiguity, feasibility concerns, and context dependencies. Where comments indicate unclear wording, the requirement statement is revised while preserving traceability to the original ID and rationale.

12.3.3 Consensus reporting (IQR-based only)

Consensus is reported using the Interquartile Range (IQR) of expert ratings for each requirement. Requirements are categorised as:

- **High consensus:** $IQR \leq 1.0$

- **Moderate/partial consensus:** IQR = 1.0–2.0
 - **Low consensus:** IQR > 2.0
- These thresholds are applied consistently across the 7-point scale to support transparent prioritisation and to identify requirements requiring clarification or re-specification during WP5 refinement.

12.4 Questionnaire (The full instrument)

DEL-R01 (Q01). Protection against cyber threats and robust privacy safeguards (including incident response and continuity plans) are fundamental preconditions for deploying an antifragile urban mobility decision support platform.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R02 (Q02). PESTEL factors (political, economic, social, technological, environmental, legal) should be explicitly integrated into the system's decision rules and tradeoff visualisations.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R03 (Q03). Predefined, transparent evacuation and rerouting priority schemes, including provisions for people with special mobility needs, must be embedded in the system for disruptive transport events.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R04 (Q04). Mechanisms for sharing, benchmarking and validating resilience and antifragility practices across cities should be embedded in the platform to support cross-city learning.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R05 (Q05). Resilient densification strategies (e.g. compact high demand hubs, peripheral sub-centres, 15minutecity principles) should be explicitly modelled in the mobility decision support system to enhance antifragility.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R06 (Q06). For an antifragile urban mobility system, public transport, walking and cycling must be represented with equal data coverage and analytical weight as private motorised modes in all core functions (monitoring, prediction, routing).

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R07 (Q07). The system should support rapid end-to-end response to major events (from detection to deployment of measures) within a few minutes, for example less than 2 minutes for high criticality incidents.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R08 (Q08). Shared situational awareness platforms and predefined coordination protocols are required to support multi stakeholder decision-making during transport disruptions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R09 (Q09). Signal control and routing logic should jointly optimise resilience and environmental performance (e.g. reducing idling and emissions) rather than treating environmental gains as secondary co-benefits.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R10 (Q10). For large scale transport disruptions, a centralised command view with one click activation of predefined flow redirection and inter-agency coordination protocols is required.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R11 (Q11). An antifragile mobility system requires live traffic data for all major links, with updates at intervals not exceeding 5 minutes and embedded anomaly detection.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R12 (Q12). User interfaces must provide accurate, accessible and multilingual information tailored to different user groups, including during crisis situations.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R13 (Q13). The platform should track how a disruptive event in one urban subsystem (e.g. rail, power) propagates across other systems and transport modes, capturing cascading effects.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R14 (Q14). Realtime interactions and dependencies between transport modes must be modelled to prevent cascading failures and to support coordinated multimodal interventions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R15 (Q15). AI/ML based predictive maintenance of transport infrastructure should be integrated to enable preemptive repairs before disruptions occur.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R16 (Q16). Coordination across governance levels (local, regional, national) should be supported through shared dashboards and policy synchronisation tools to avoid conflicting actions in transport crises.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R17 (Q17). Continuous 24/7 monitoring of mobility conditions with anomaly detection and comparison to historical baselines is necessary to detect weak signals and learn from disruptions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R18 (Q18). Routing and evacuation functions should systematically incorporate individual mobility constraints (e.g. disability, age, temporary health restrictions) when generating recommendations during disruptions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R19 (Q19). The system should use user preferences and past disruption responses (within privacy constraints) to nudge travellers towards more sustainable choices such as public transport, cycling, off-peak travel and carpooling.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R20 (Q20). Context specific qualitative severity and impact scales, codesigned with stakeholders, should be applied consistently across all system outputs for transport disruptions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R21 (Q21). Data from multiple transport modes and sources (sensors, operators, crowdsourcing) should be fused into a unified analytical framework, with reconciliation of inconsistencies between sources.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R22 (Q22). Recording data provenance and trust scores for critical inputs (e.g. crowdsourced incident reports) is necessary to support credible and transparent decision-making.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R23 (Q23). Edge computing nodes capable of maintaining essential control functions during central system or communication failures are required to avoid systemic collapse under severe transport disruptions.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R24 (Q24). Economic cost and distributional impact modelling of disruptions and interventions should be integrated into the decision support environment for urban mobility.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R25 (Q25). Regulatory and legal constraints, including emergency powers and temporary orders, should be encoded and kept up to date in the system's decision logic.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R26 (Q26). The system should calculate and track antifragility metrics (e.g. post disruption performance improvements) over time, with possibilities for benchmarking across cities or subsystems.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R27 (Q27). The system should support long term institutional learning, including signalling when transformative (rather than incremental) changes in urban mobility are warranted.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R28 (Q28). Maintaining and prioritising mobility for emergency services (e.g. via signal preemption and protected routes) should be an explicit objective under all disruption scenarios.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R29 (Q29). For time critical incidents, decisionmakers should receive comparative scenario evaluations (e.g. within 5 minutes) before selecting an intervention.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R30 (Q30). Role based access control and data partitioning should enable different entities to collaborate securely within a defined trust framework (usability across split governance).

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R31 (Q31). The system should demonstrably improve performance (e.g. throughput, delays, recovery time) when exposed to repeated disruptions of increasing intensity, thereby exhibiting antifragility.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R32 (Q32). Incident management should rely on dynamic prioritisation based on severity, escalation potential and impact on vulnerable populations, harmonised with the agreed scales.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R33 (Q33). A graph based data model of the urban mobility network (nodes, edges, attributes) should underpin routing, disruption impact analysis and real-time topology updates.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R34 (Q34). Authorised stakeholders should be able to inject new information (e.g. local constraints, emergent risks) into the system via secure mechanisms with full audit trails.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R35 (Q35). Transport planning and emergency routing algorithms must incorporate geographic and environmental constraints (e.g. terrain, protected areas, flood zones, climate hazards) as first class inputs.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R36 (Q36). The system should predict emerging disruptions and proactively reroute traffic before congestion or blockages fully develop, updating its models based on observed outcomes.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R37 (Q37). Ensuring equitable access for underserved groups and avoiding disproportionate disadvantages during disruptions should be a core design principle of the mobility system.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R38 (Q38). A comprehensive set of resilience and antifragility KPIs linked to policy objectives should be monitored via realtime dashboards, periodic reports, and threshold alerts.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R39 (Q39). Realtime dashboards should monitor how mobility conditions are distributed across geographic areas and demographic groups, issuing alerts for disproportionate impacts.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R40 (Q40). During disruptions affecting motorised transport, the system should be able to prioritise and actively promote walking and cycling as primary alternatives where feasible.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R41 (Q41). Scenario simulation tools for unexpected events (including compound or rare shocks) are necessary for stress testing, operator training, and preparedness exercises.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R42 (Q42). Vehicle routes and fleet allocation should be dynamically adjustable during disruptions without relying solely on pre-scheduled contingency plans.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R43 (Q43). Ex-ante and ex-post evaluations of transport measures and projects on resilience and antifragility KPIs should be systematically performed and fed back into policy cycles.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R44 (Q44). Guidance functions should optimise for people moved (individual mobility) rather than vehicle throughput, including dynamic reallocation of demand to walking and cycling when conditions allow.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R45 (Q45). The system should respect user autonomy by offering a set of viable routing and modal options aligned with user preferences, rather than enforcing a single solution.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R46 (Q46). During both normal and disrupted conditions, control strategies should actively minimise energy use and emissions, for example via adaptive signal timing that reduces average vehicle idling.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree

DEL-R47 (Q47). Decision-support tools for projecting likely outcomes of disruption scenarios or intervention options, including policy impact and evolving crisis forecasting.

☐ 1 = Strongly disagree ☐ 2 = Disagree ☐ 3 = Slightly disagree ☐ 4 = Neither agree nor disagree ☐ 5 = Slightly agree ☐ 6 = Agree ☐ 7 = Strongly agree